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BIOLOGY 1 ASB0204

Topic 5
Plant and Animal Tissues



Biology Unit
Centre of Foundation Studies for Agricultural Science
Universiti Putra Malaysia

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Learning outcomes

By the end of this video lecture, you will be able to:

1. Describe the types of plant and animal tissues (C2)
2. Differentiate types of plant and animal tissues (C3)

Outline

5.1 Cells and Tissues of Flowering Plants (24.1)

5.2 Types of Animals Tissues (31.1)

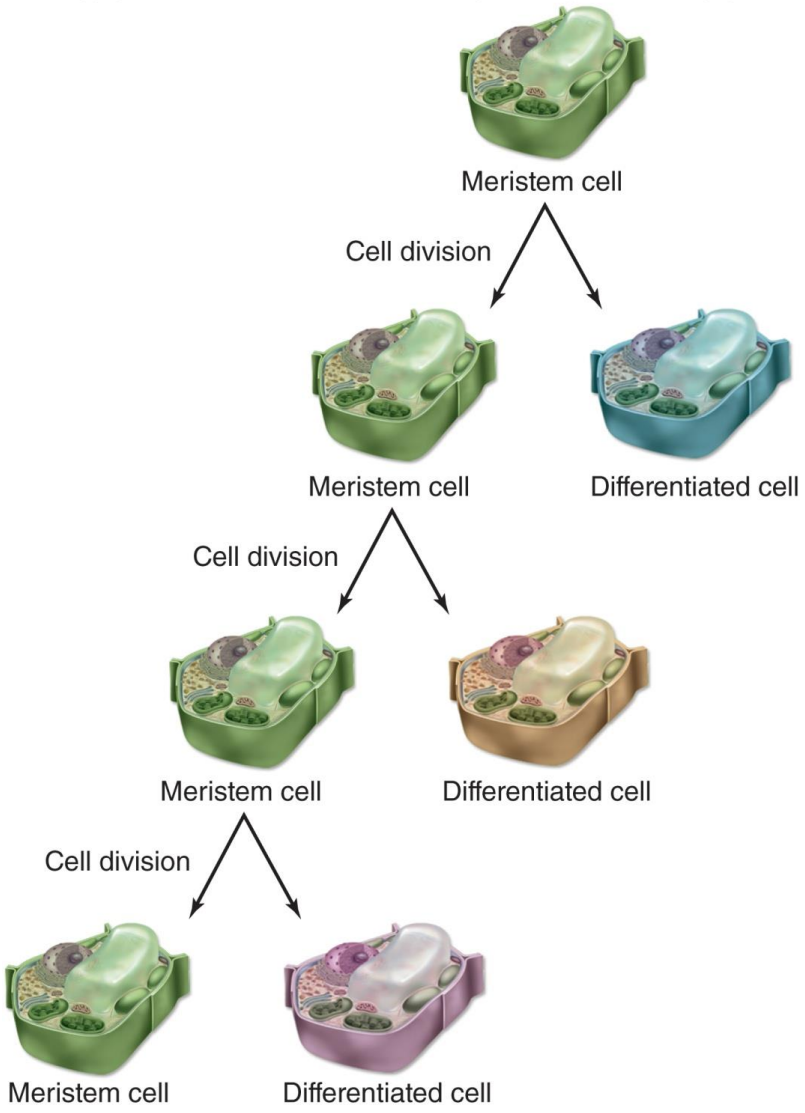
*(X.X) refers to chapter in TEXTBOOK

5.1 Cells and Tissues of Flowering Plants

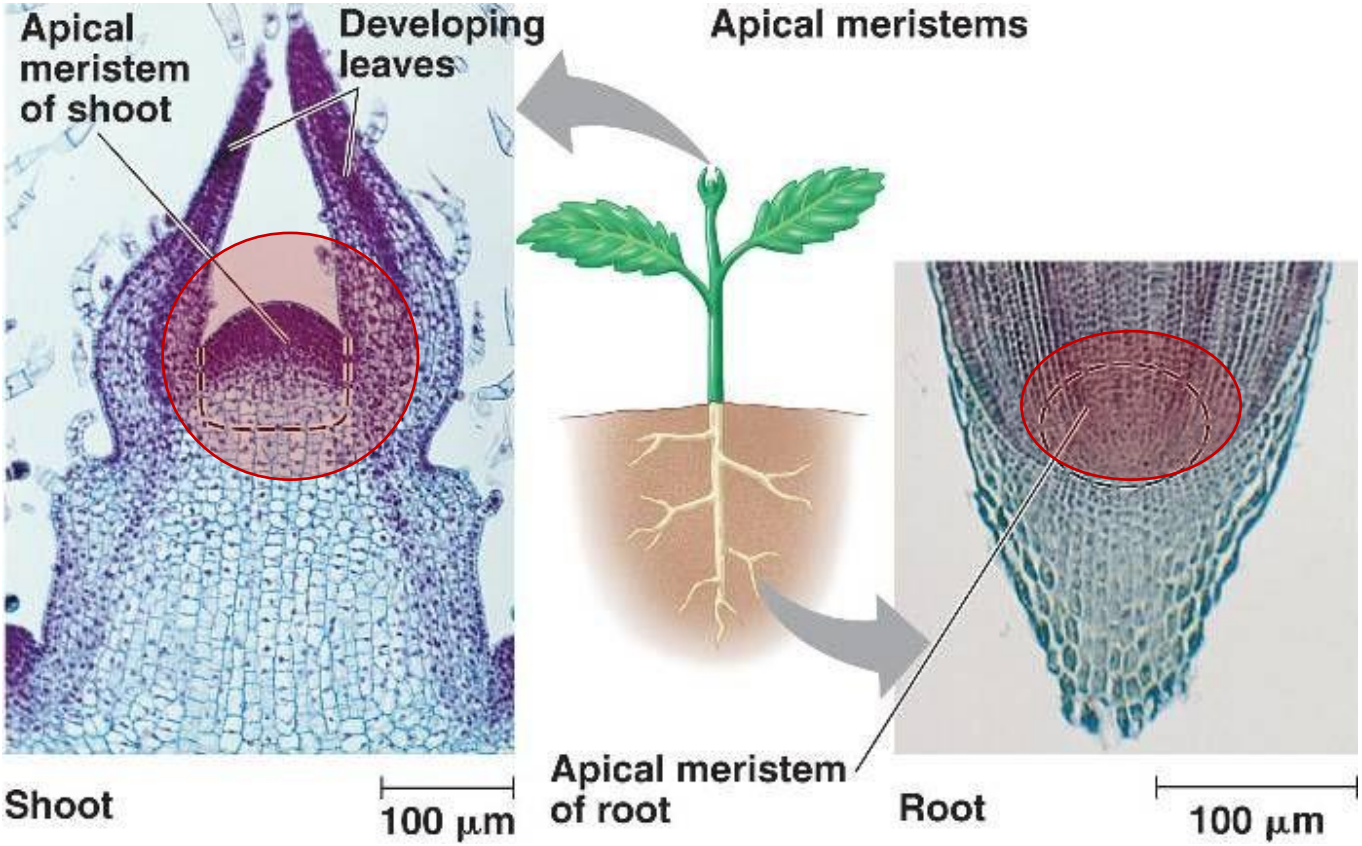
- As plant embryo begins to develop – the first cell are called **meristem cells**
  - ❑ Meristematic cells – **undifferentiated** cells, able to **divide indefinitely** and give rise to many types of **differentiated cells**

- Apical meristems at the of stems and roots increase the length of these tissues

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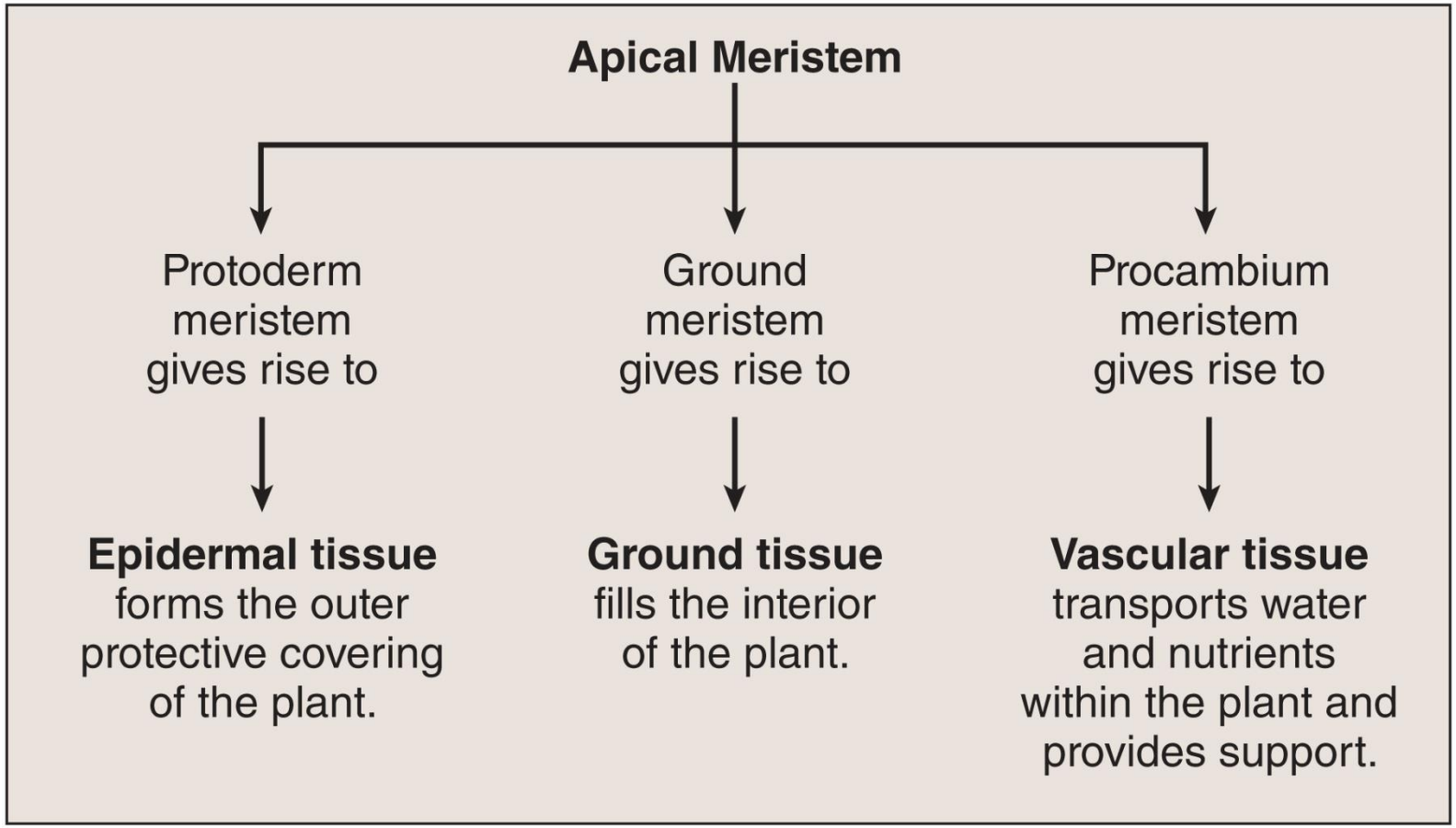
- Meristem cells present at the tips or apices of the young plant : **apical meristem of shoot** and **apical meristem of root**



https://www.pinterest.com/pin/395613148494365071/

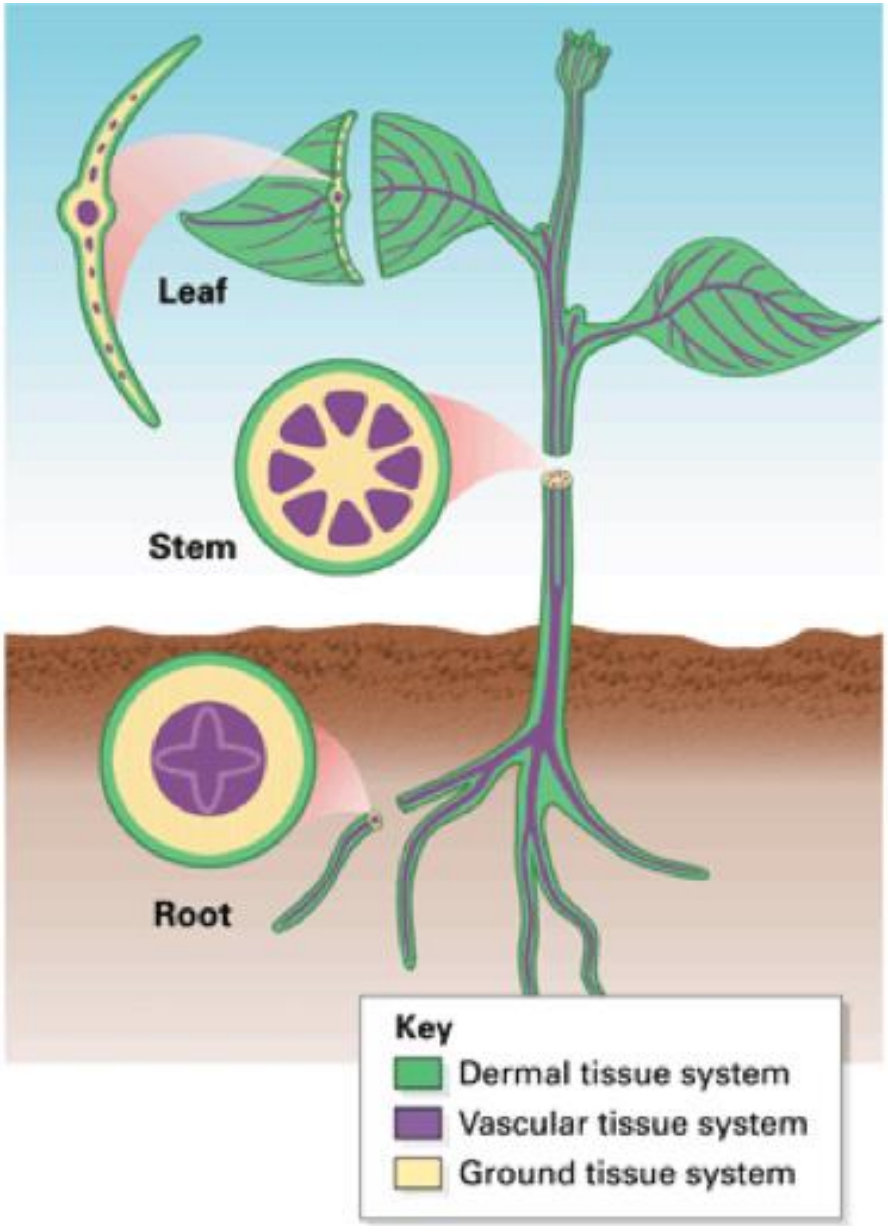
- Apical meristems produce three types of **meristem**, which produce three specialized tissues; **Epidermal**, **Ground**, **Vascular Tissues**

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Types of Plant Tissue Systems;

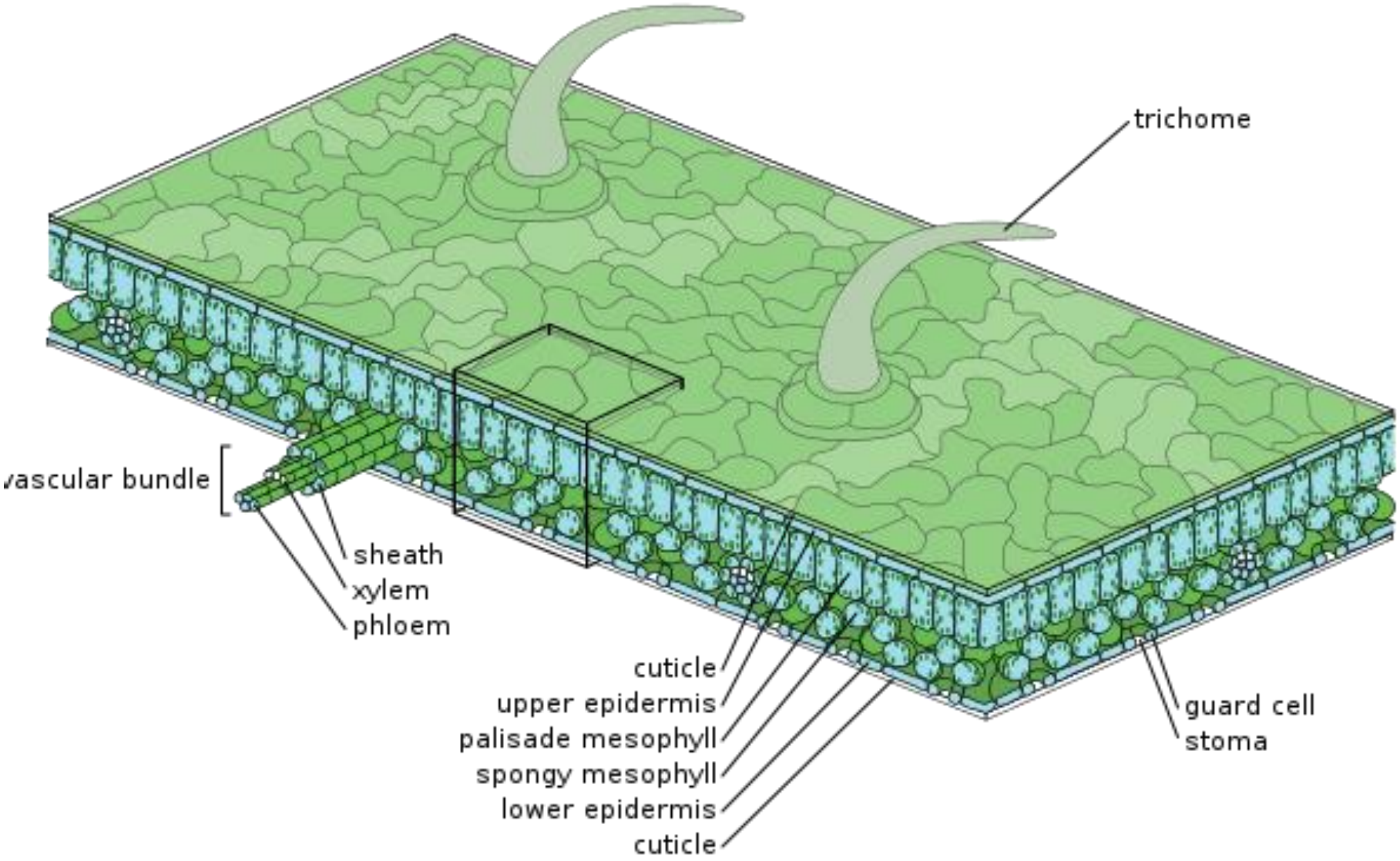
1. Epidermal/Dermal Tissue System
2. Vascular Tissue System
3. Ground Tissue System



http://eima86.tripod.com/id2.html

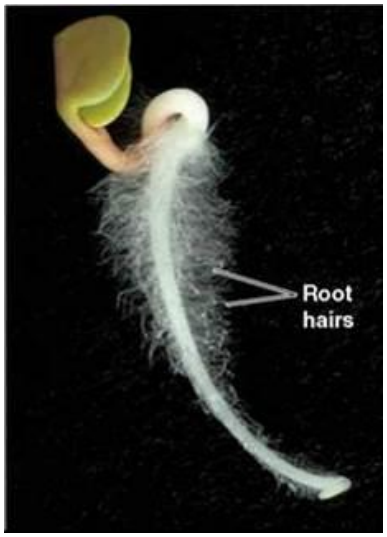
Plant Tissue System: Dermal Tissue

- Entire body of woody and non-woody plants contains closely packed epidermal cells called **epidermis**
- Forms the outer **protective covering** of a plant
- Epidermis;
  - ❑ Some are covered with **waxy cuticle**
  - ❑ Root epidermal cells have **root hairs** – increase surface area for water absorption
  - ❑ Stem, leaves and reproductive organs have **trichomes** (protect the plants from too much sun and moisture loss, protect plant from herbivore by produce toxic substances)
  - ❑ Lower leaf surface contains **stomata** for gas exchange


trichome
vascular bundle
sheath
xylem
phloem
cuticle
upper epidermis
palisade mesophyll
spongy mesophyll
lower epidermis
cuticle
guard cell
stoma



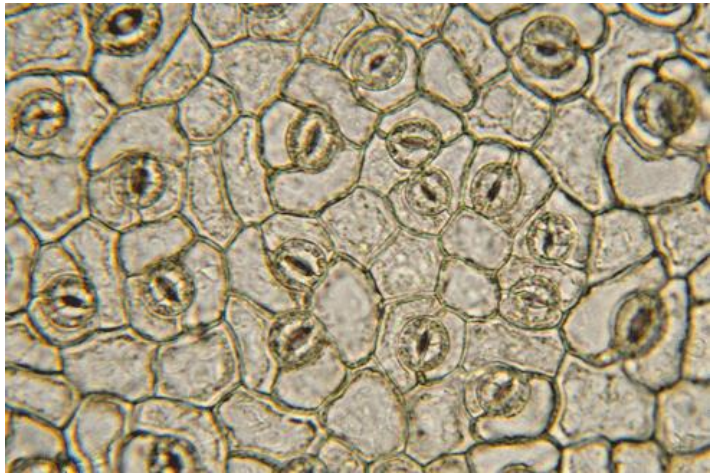
cuticle



root hairs



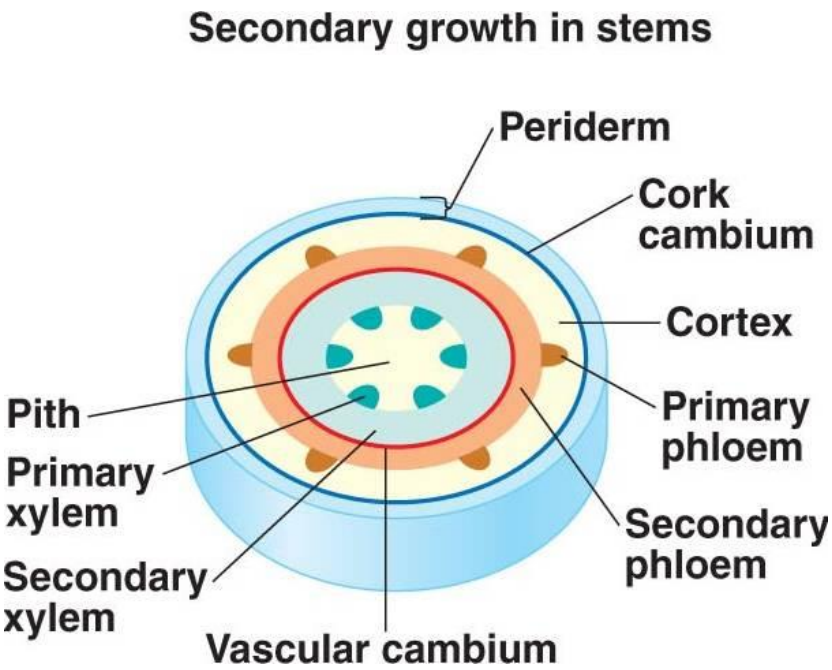
trichomes



stomata

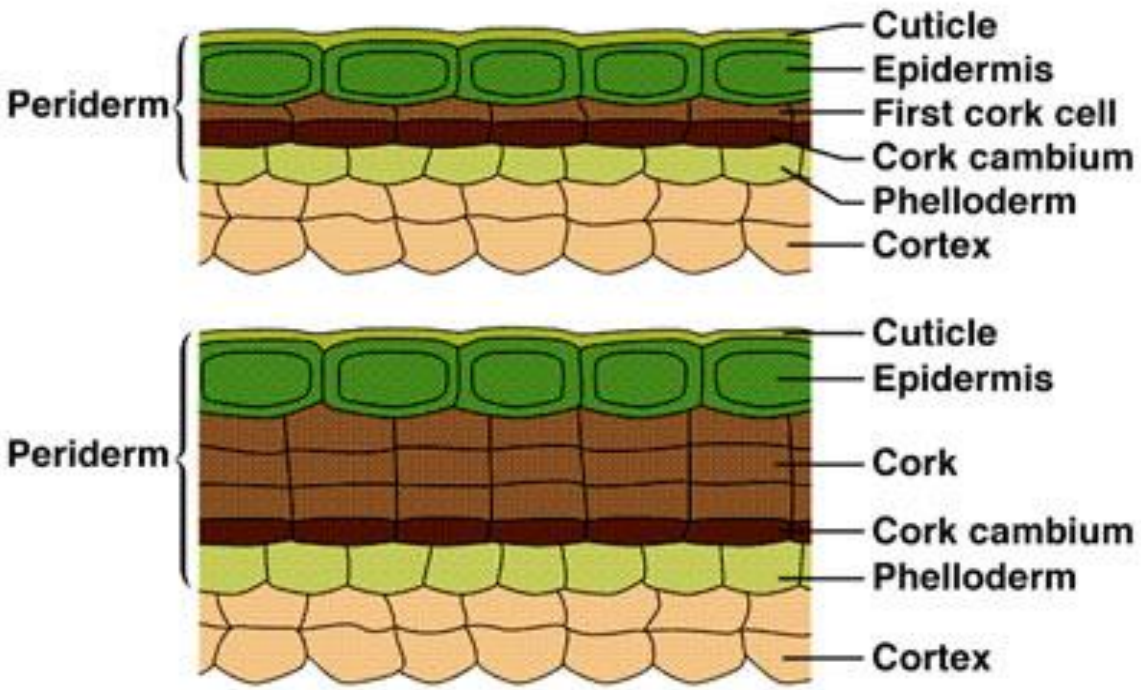
Plant Tissue System: Dermal Tissue

- In older woody plants, the epidermis of the stem is replaced by periderm (cork) during **secondary growth** of plants
  - ❑ Major component contains **cork cells**
  - ❑ New cork cell is made by **cork cambium**
  - ❑ **Periderm** (nonliving) – protect the plants, resists fungal infection, bacteria and animal attacks



http://bot1320.nicerweb.com/Locked/media/ch10/growth-secondary.html

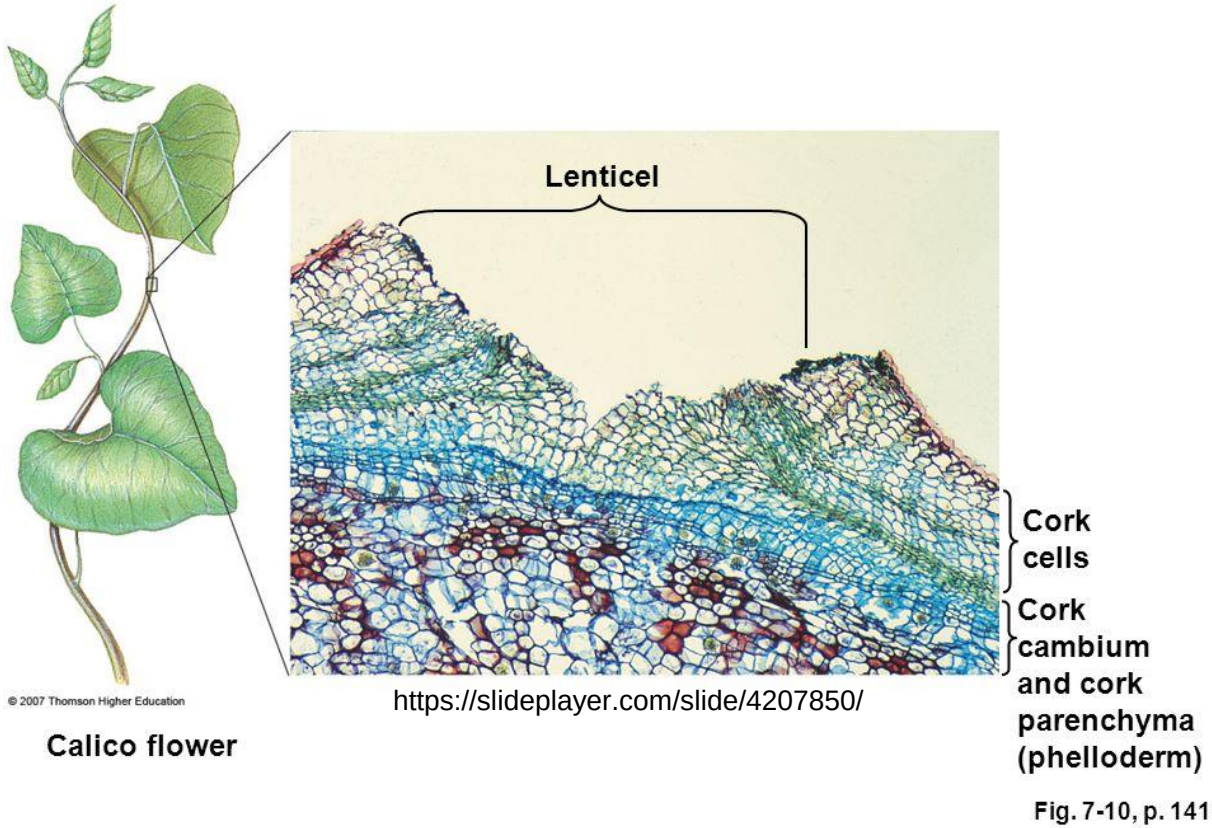
Plant Tissue System: Dermal Tissue



https://www.qsstudy.com/biology/formation-periderm-plants

- As the new cork cells mature – slightly increase in girth, walls become encrusted with **suberin (lipid)**, they are waterproof and chemically inert
  - ❑ Protect the plant cells to resist fungal, bacteria and animal attacks.

Plant Tissue System: Dermal Tissue



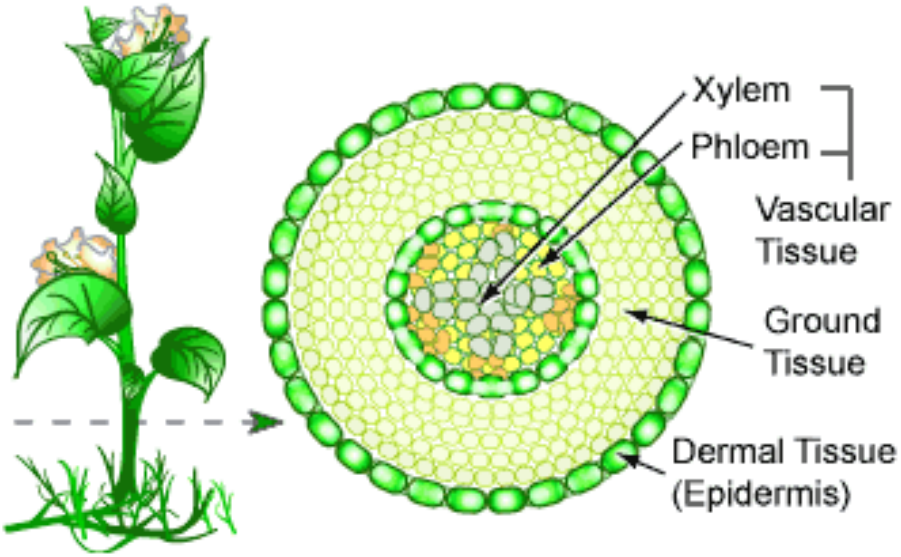
https://slideplayer.com/slide/4207850/

Fig. 7-10, p. 141

- **Lenticel** – overproduces cork by cork cambium in the certain areas of stem surface (causing ridges and cracks).
  - ❑ site of gas exchange between interior of a stem and the air

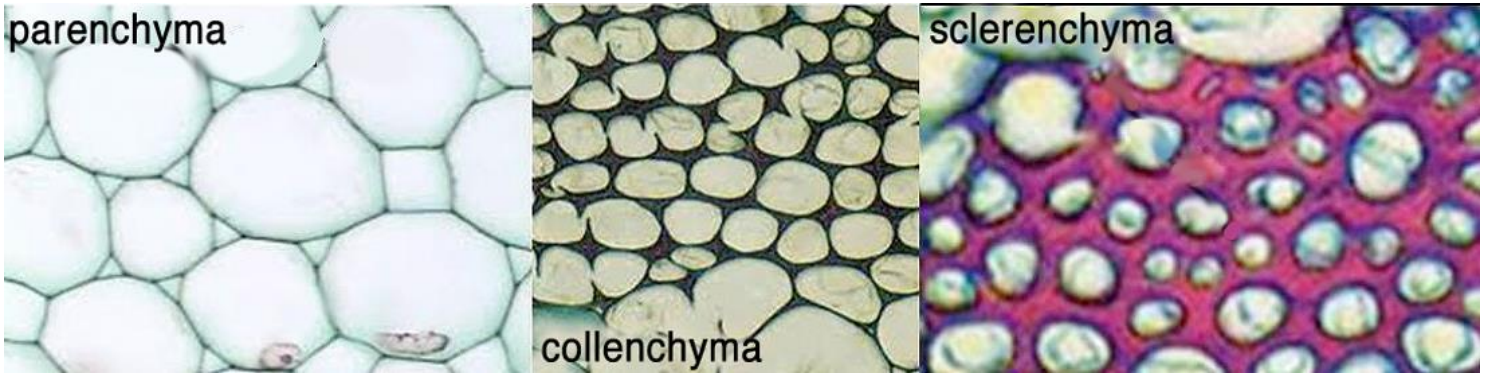
Plant Tissue System: Ground Tissue

- Ground tissue **forms bulk** of a flowering plants
- Cells between dermal and vascular tissues
- **Functions:** provides support and storage of materials in plants



https://sbi3uiplantkingdom.weebly.com/week-12/three-major-types-of-plant-tissues

- Ground tissue composed of ; **parenchyma**, **collenchyma** and **sclerenchyma**

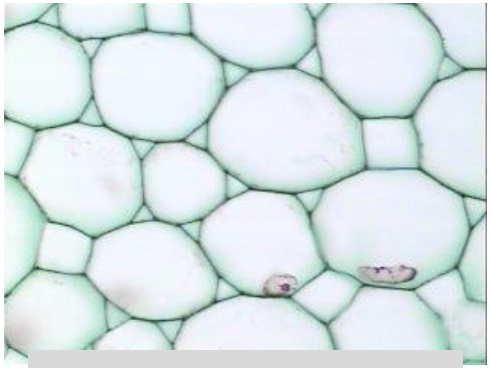


http://www.bio.miami.edu/dana/226/226F09_5.html

Plant Tissue System: Ground Tissue

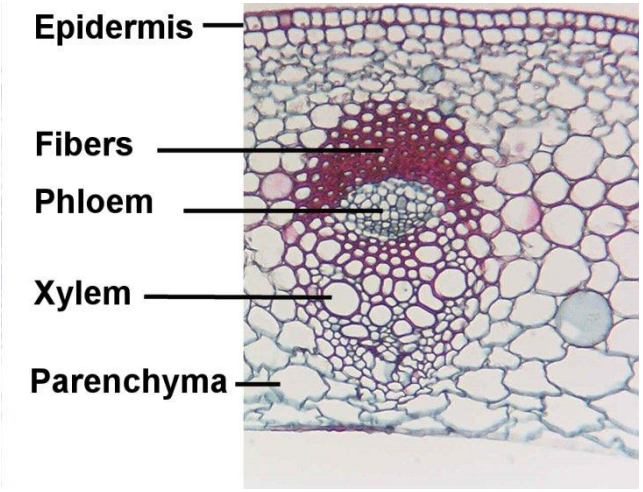
Parenchyma

- **Least specialized** and **most abundant** cells; found in all organs of plant.
  - ❑ contain **chloroplast** and carries on photosynthesis
  - ❑ contain colorless plastid to store product of photosynthesis
- Can divide and give rise to more **specialized cells**
  - ❑ e.g root develop from stem cutting placed in water
- Contain chloroplasts to perform photosynthesis and large vacuole to store water, starch and oils



Parenchyma cells

http://eima86.tripod.com/id4.html

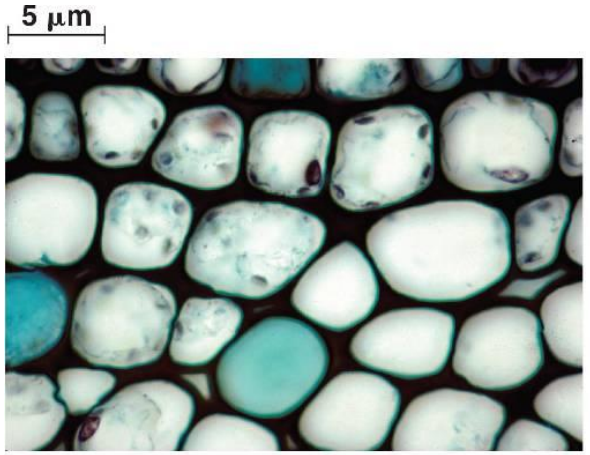


https://oplas.co.tz/question/what-is-a-parenchyma-cells/

Plant Tissue System: Ground Tissue

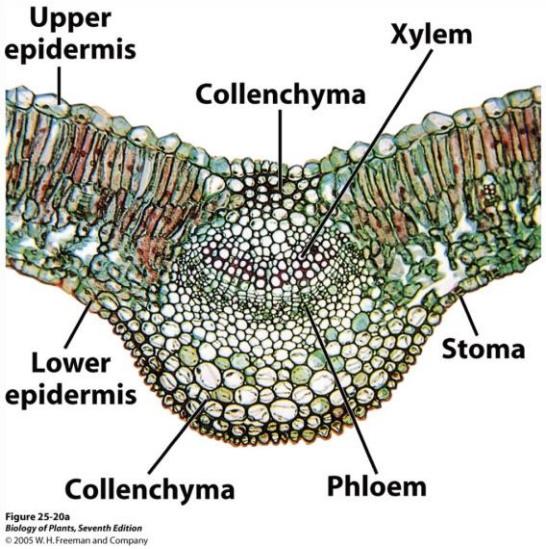
Collenchyma

- Have **thicker primary walls**
- **Form bundles** underneath epidermis:
  - ❑ Elongated cells
- Contain **no chloroplast** and present in any part of plants that is still growing
- **Main function:**
  - ❑ **provide flexible support** to young plant parts



Collenchyma cells (in *Helianthus* stem) (LM)

https://www.simply.science/images/content/biology/plant_form_and_function/growth_and_development/conceptmap/Collenchyma.html

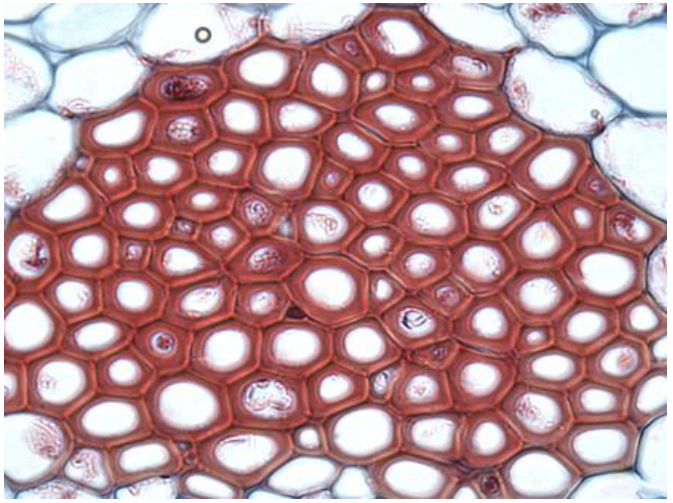


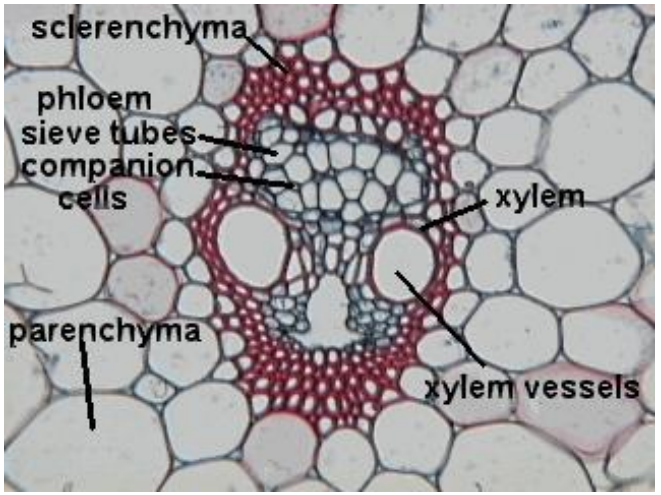
https://quizlet.com/388111434/lilac-leaf-diagram/

Plant Tissue System: Ground Tissue

Sclerenchyma

- Have **thick secondary walls** impregnated with **lignin**
- **Nonliving at maturity**: they die after they differentiate
- Scattered throughout parenchyma cells;
  - ❑ e.g give guava their gritty texture
- **Main functions:**
  - ❑ support mature regions of the plants
  - ❑ support vascular tissue
  - ❑ important component of wood



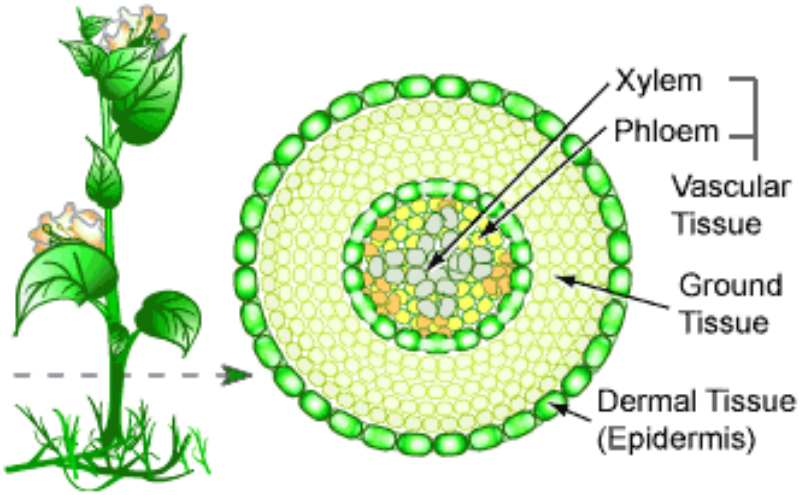


Cross section of sclerenchyma fibers

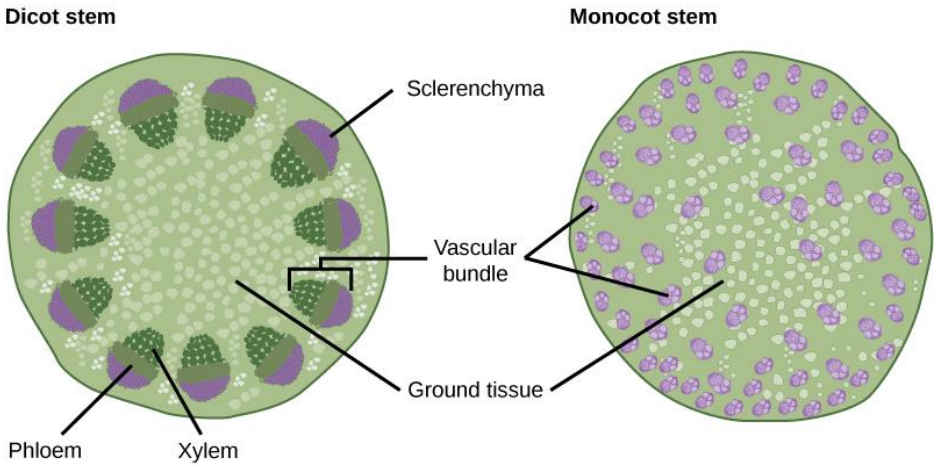
https://en.wikipedia.org/wiki/Ground_tissue

Plant Tissue System: Vascular Tissue

- Two types of vascular tissues; **xylem and phloem**
  - ❑ Xylem **transports water and minerals** from roots to the leaves
  - ❑ Phloem **transports sucrose and organic compounds** from the leaves to the root



- In root, vascular tissue located in the **vascular cylinder**; in stem vascular tissue located in the **vascular bundles**; in leaves in found in the **leaf veins**



Plant Tissue System: Vascular Tissue

Xylem

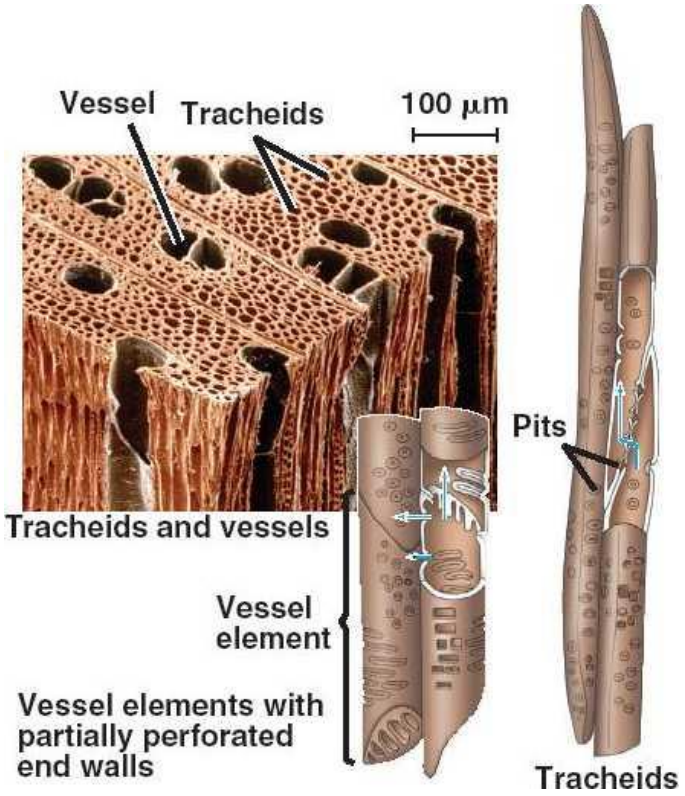
- Water and minerals conducting system, made of dead cells with lignified walls
- Roots – found in vascular cylinder; stem – in vascular bundles; leaves in leaf veins
- Types of conducting cells:

Tracheid

- ❑ long and tapered ends
- ❑ water moves across pits in the end walls and side walls

Vessel elements

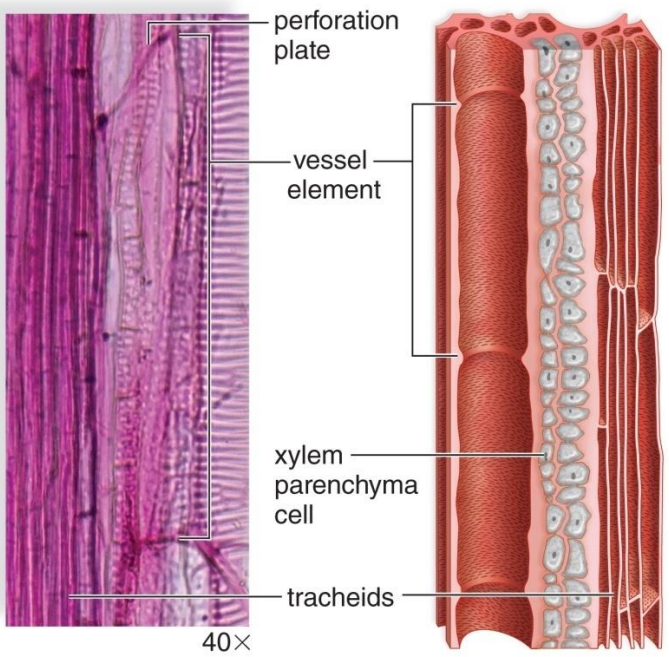
- ❑ larger with perforation plates in their end walls
- ❑ form a continuous vessel for water and mineral transport



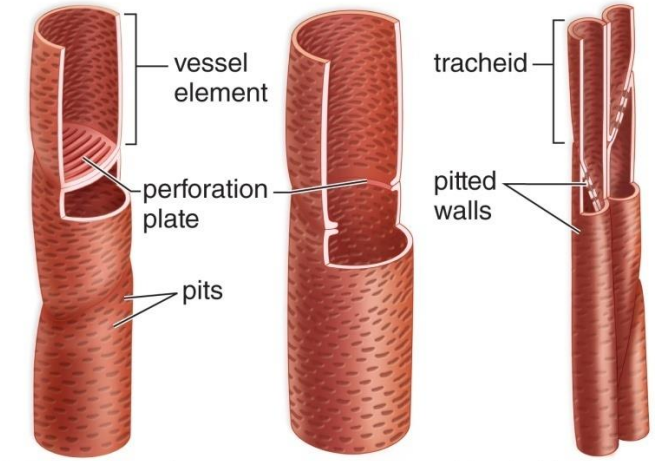
http://bio1151b.nicerweb.net/Locked/media/ch35/plantcells_xylem.html

Xylem Structures

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a. Xylem photomicrograph (*left*) and drawing (*to side*)



b. Two types of vessels c. Tracheids

a: © Garry DeLong/Science Source

Plant Tissue System: Vascular Tissue

Phloem

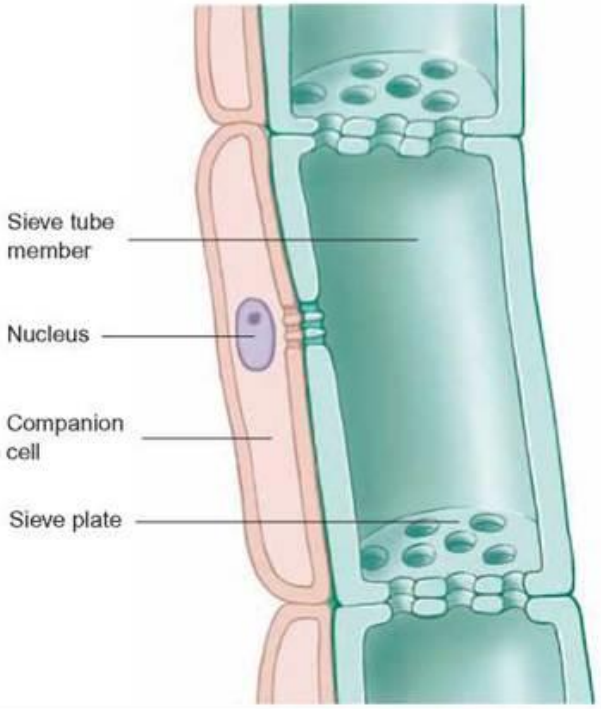
- Transport sucrose and other organic compound from the leaves to the roots
- Conducting cells of phloem : sieve tube members and companion cells

Sieve tube members

- ❑ Arranged end to end to form sieve tube
- ❑ Each sieve-tube member has a companion cell that contain nucleus

Companion cells

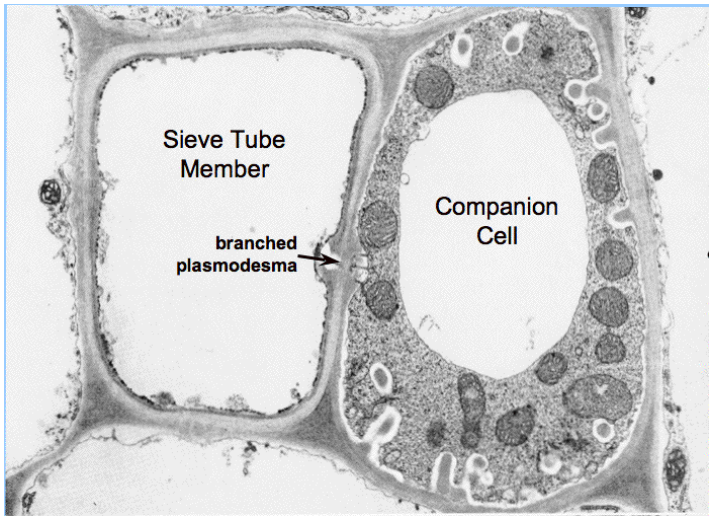
- ❑ Closely connected to sieve tube members by plasmodesmata
- ❑ Have nucleus – control and maintain the function of both cells



https://www.google.com/search?q=phloem+sieve+tube+and+companion+cells&tbm=isch&ved=2ahUKEwjbsJ3h2orqAhUSDrcAHe6bCa8Q2-cCegQIABAA&oq=phloem+sieve+tube+and

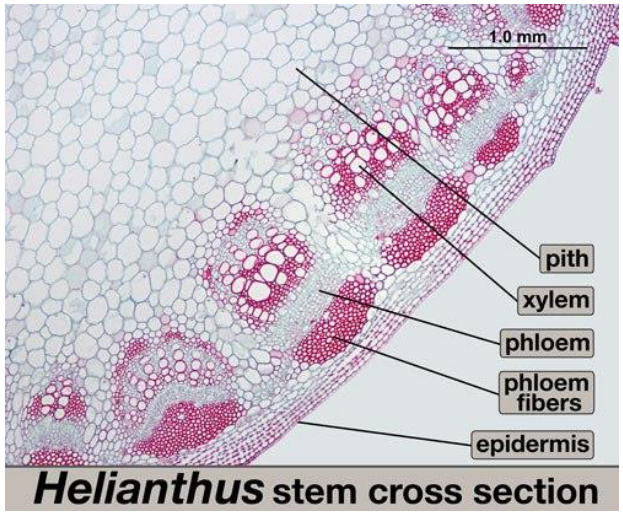
Plant Tissue System: Vascular Tissue

Phloem



https://study.com/academy/lesson/companion-cells-in-plants-function-lesson-quiz.html

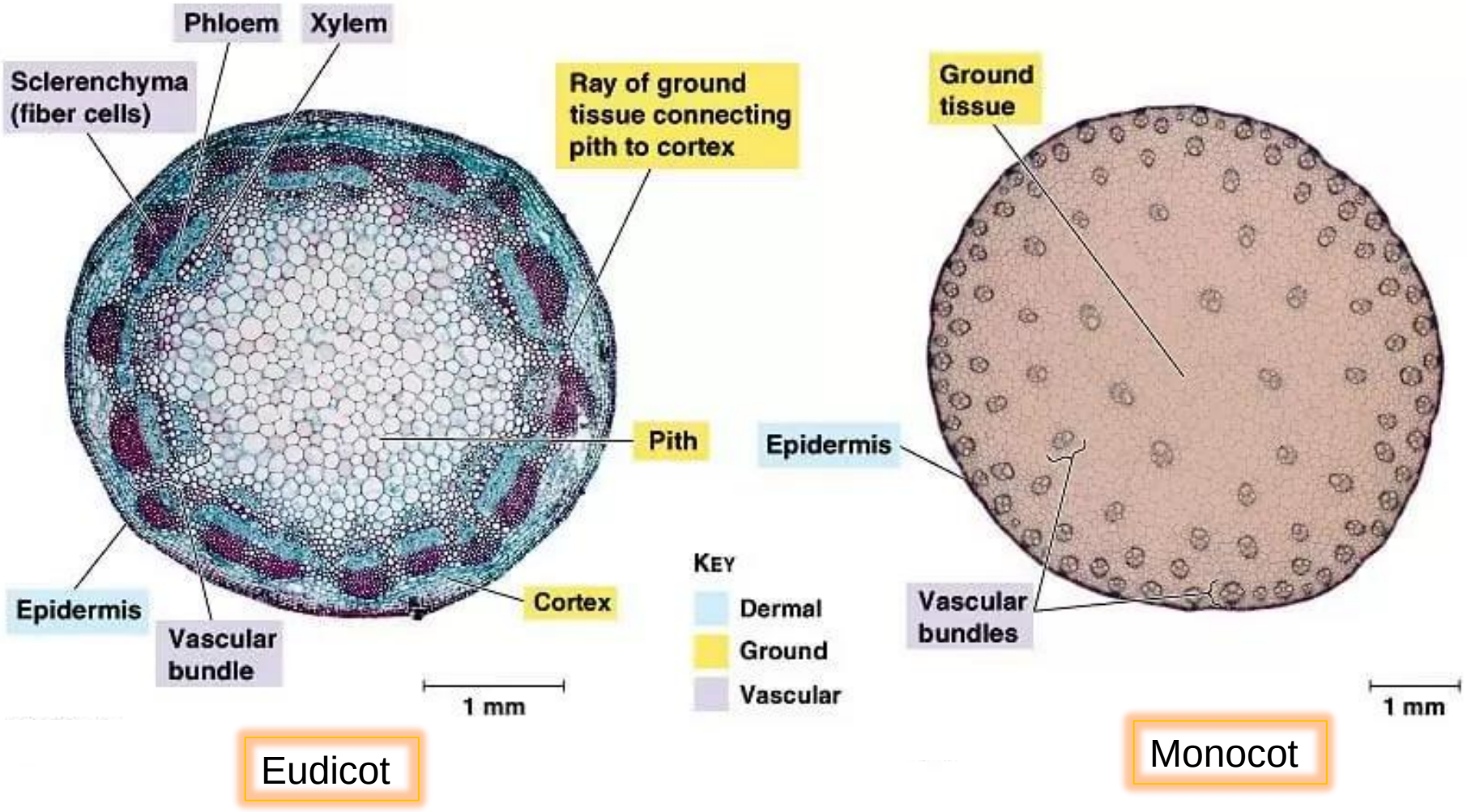
- Each sieve tube member has companion cell which are connected by plasmodesmata



https://plantbiologych9.wordpress.com/transport-in-the-phloem-of-plants/

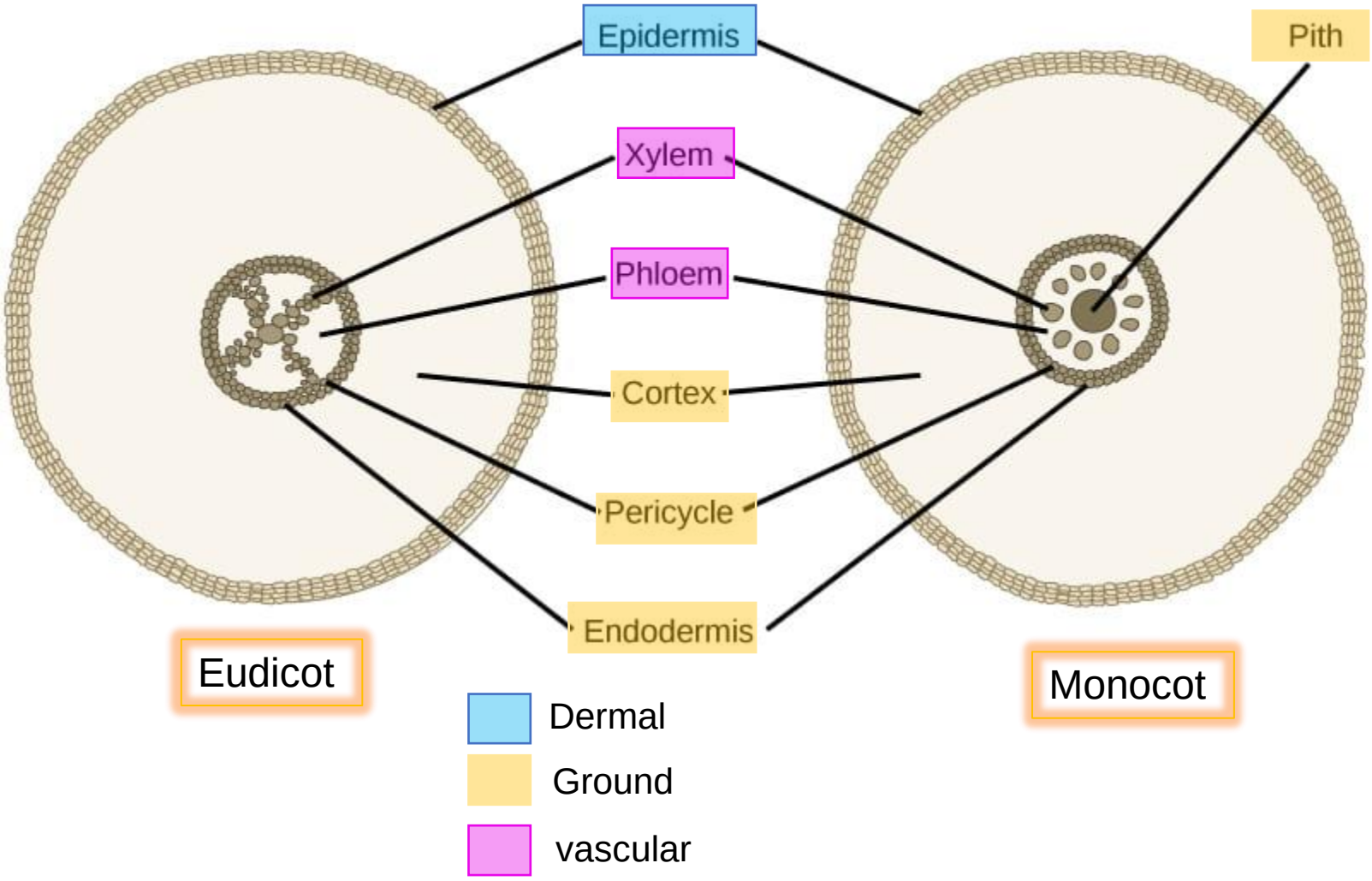
- Sclerenchyma fibers – lend support to phloem

Distribution of plant tissues in eudicot and monocot stems



https://diffzi.com/dicot-stem-vs-monocot-stem/

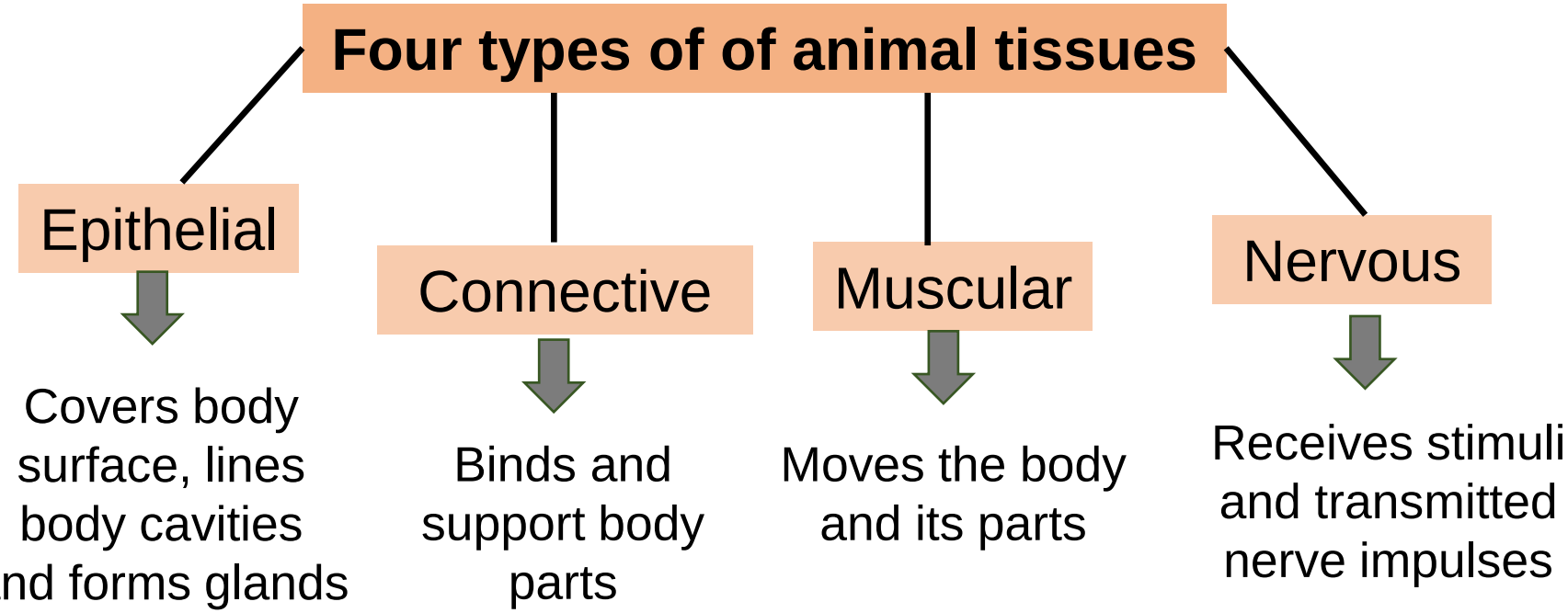
Distribution of plant tissues in eudicot and monocot roots



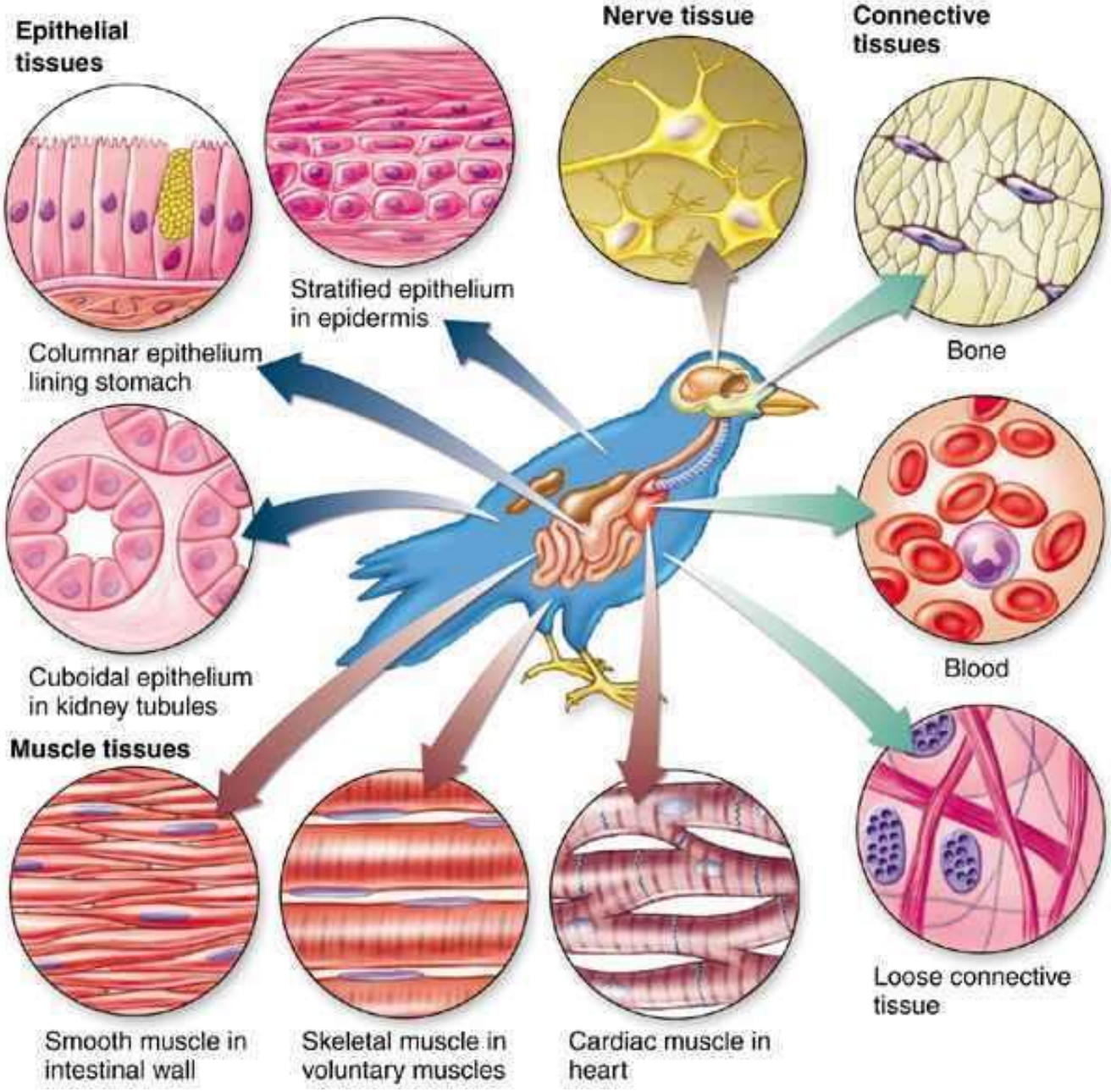
https://organismalbio.biosci.gatech.edu/growth-and-reproduction/plant-development-i-tissue-differentiation-and-function/

5.2 Types of Animal Tissue

Tissues :
collection of specialized cells of the similar type that perform a common function in the body



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Animal Tissue: Epithelial Tissue

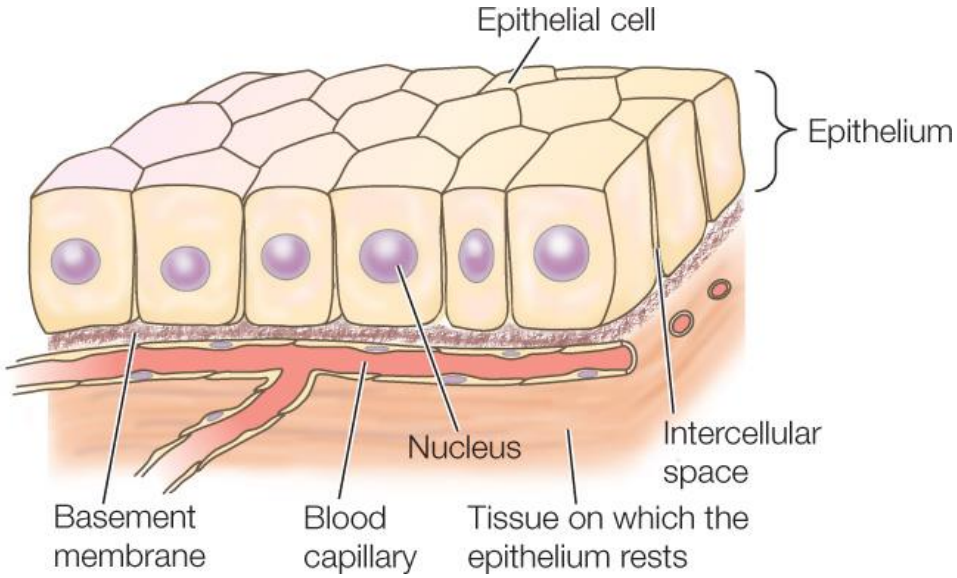
Four tissue types of animal

- **Epithelial**
  - **Covers body surface, lines body cavities and forms glands**
- Connective
  - Binds and support body parts
- Muscular
  - Moves the body and its parts
- Nervous
  - Receives stimuli and transmitted nerve impulses

Animal Tissue: Epithelial Tissue

General characteristics:

- **Tightly packed** cells that from a **continuous layer**
- Cover surface and lines **body cavities**
- Has protective function and can be modified to carry out **secretion**, **absorption**, **excretion** and **filtration**
- Can be connected by one another by **tight junction**, **adhesion junction**, **gap junctions**



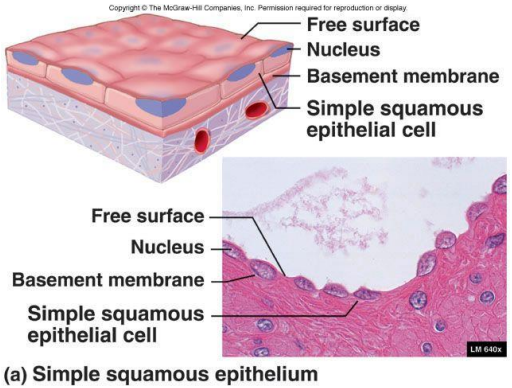
https://www.macmillanhighered.com/BrainHoney/Resource/6716/digital_first_content/trunk/test/hillis2e/asset/img_ch29/c29_fig13.html

Animal Tissue: Epithelial Tissue

- Simple epithelia – a single layer of cells and classified based on cell types

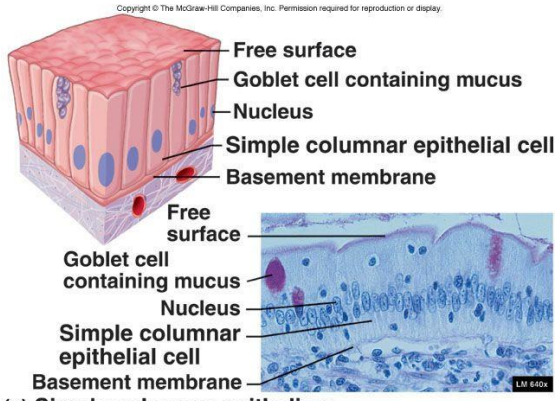
Three types of epithelial cell

Squamous epithelium – flattened cells



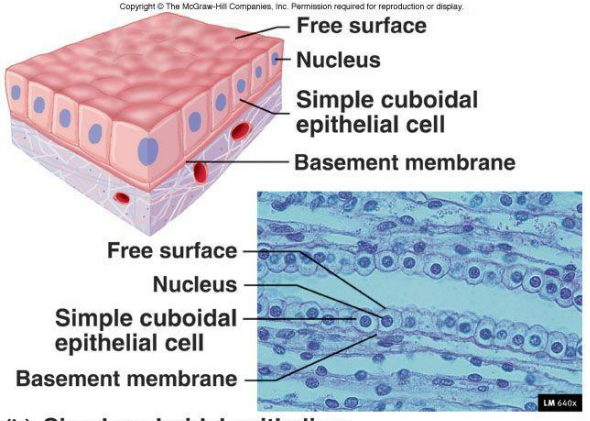
(a) Simple squamous epithelium

Columnar epithelium – cell resembling columns



(c) Simple columnar epithelium

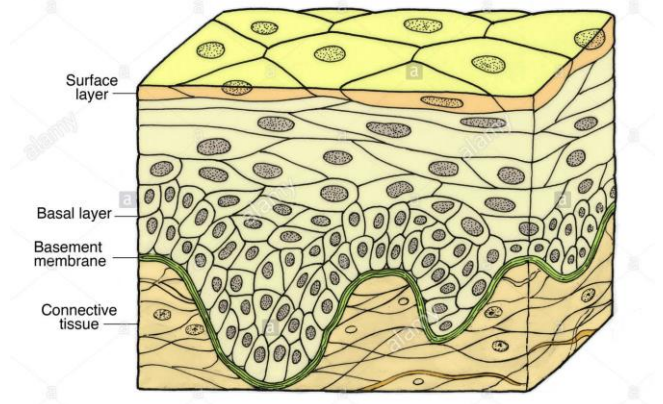
Cuboidal epithelium – cube-shaped cells



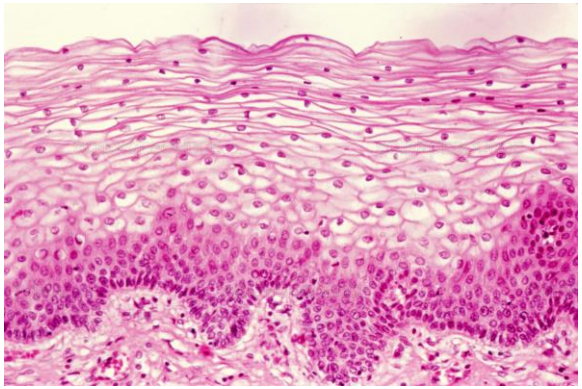
(b) Simple cuboidal epithelium

Animal Tissue: Epithelial Tissue

- **Stratified epithelia** – more than a layers of cells

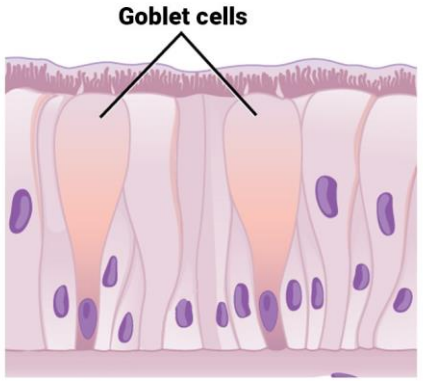


Stratified epithelium



https://www.sciencesource.com/archive/Stratified-Squamous-Epithelium-SS2181266.html

- **Glandular epithelia** – Secretes a product
  - ❑ **Gland** can be a single epithelial or contain many cells
  - ❑ **Exocrine glands** : secrete products into ducts
  - ❑ **Endocrine glands** : secrete products internally and transported by bloodstream

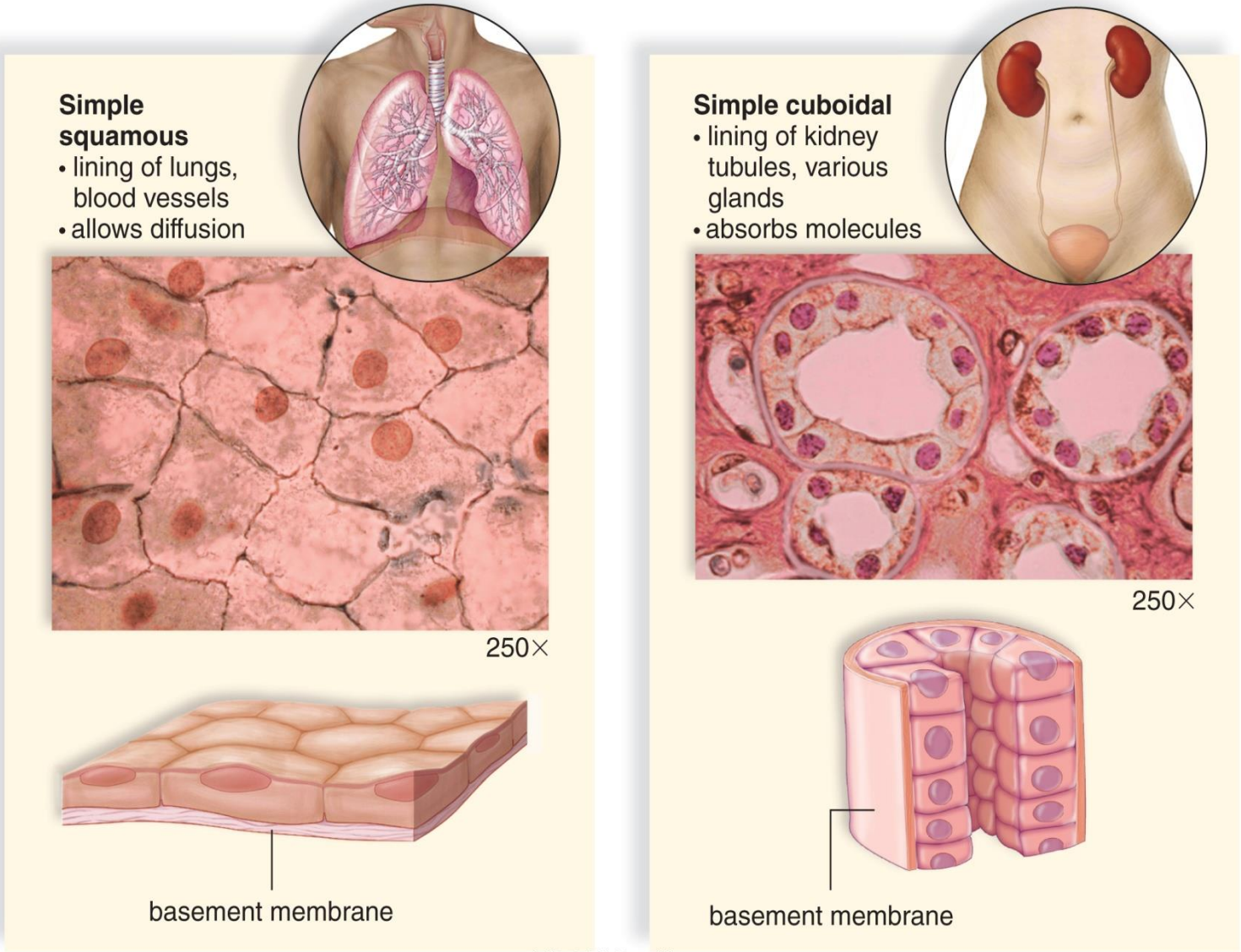


Glandular Epithelium

https://www.toppr.com/content/story/amp/types-of-epithelial-tissues-32968/

Types of Epithelial Tissue in the Vertebrates

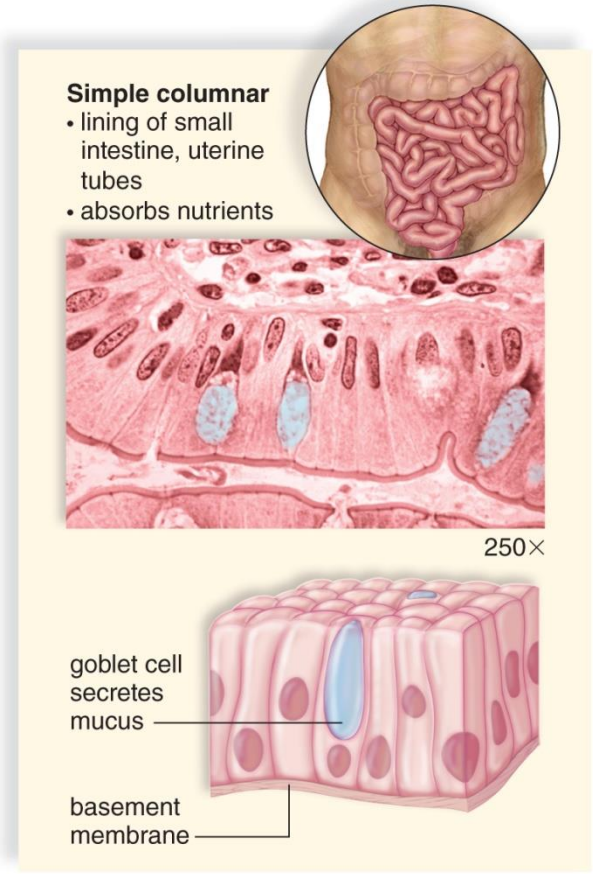
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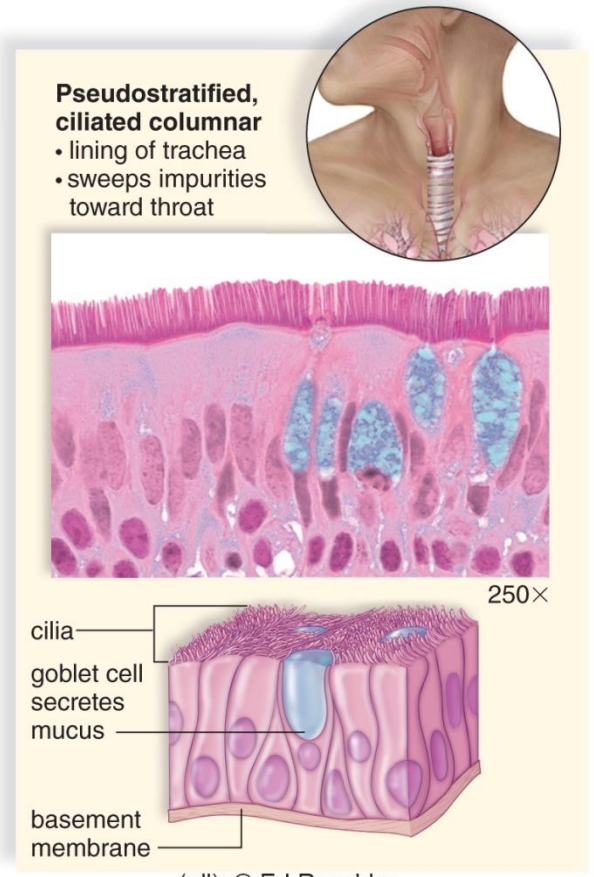


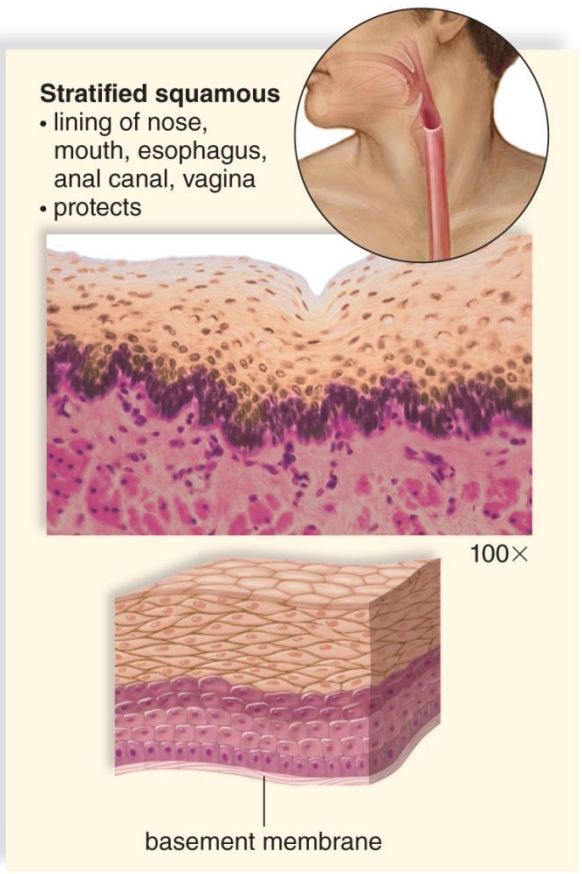
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Types of Epithelial Tissue in the Vertebrates

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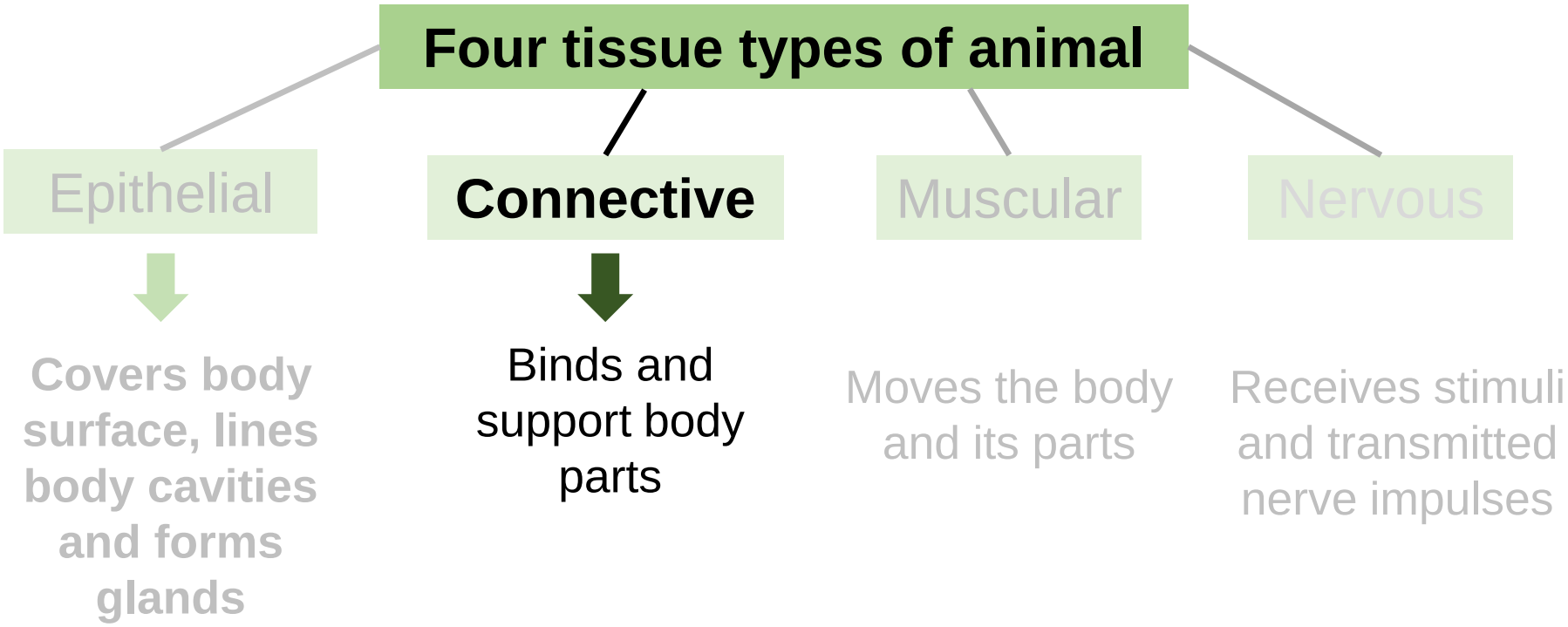




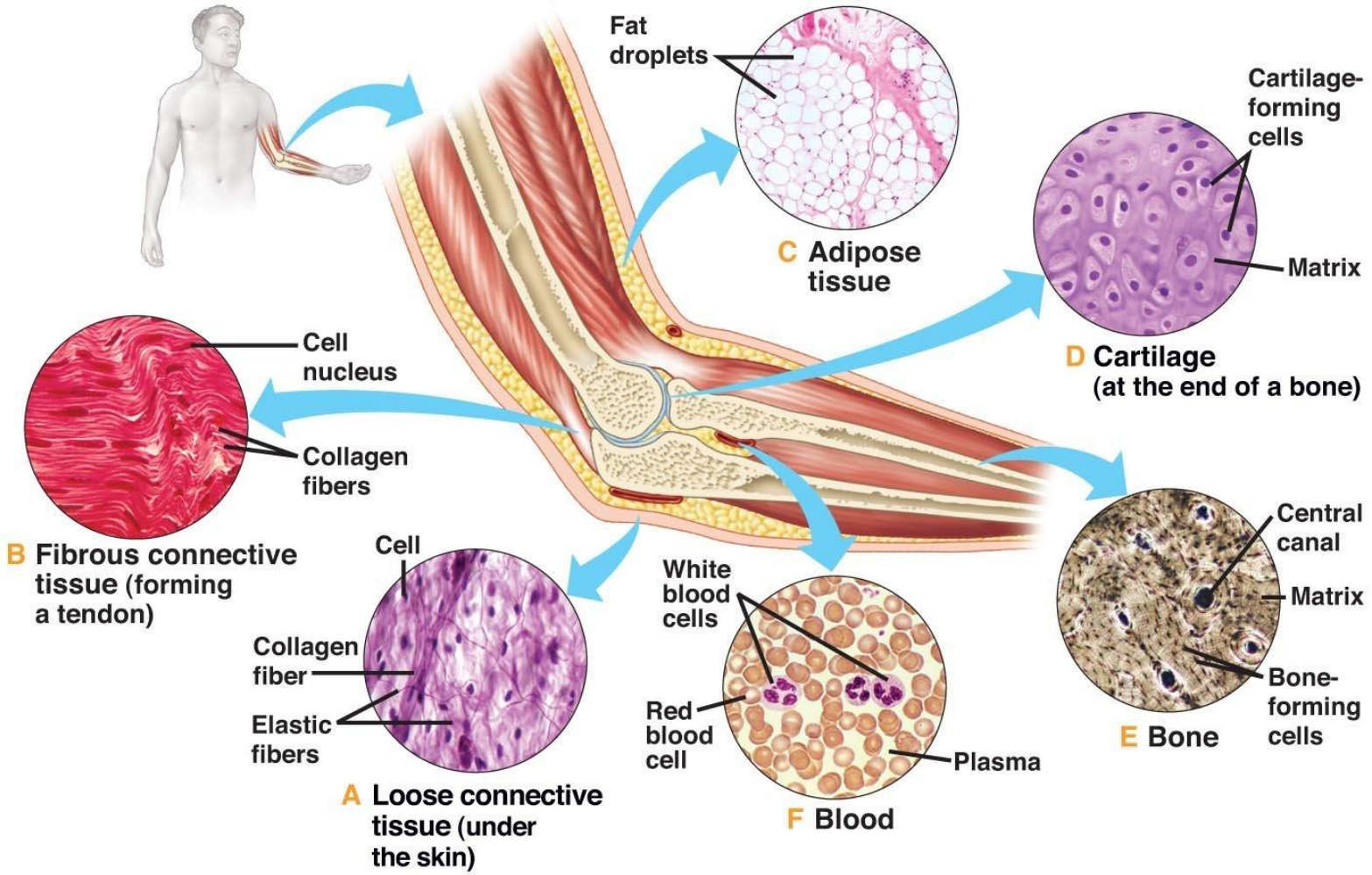


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Animal Tissue: Connective Tissue



Connective Tissues in Human Body

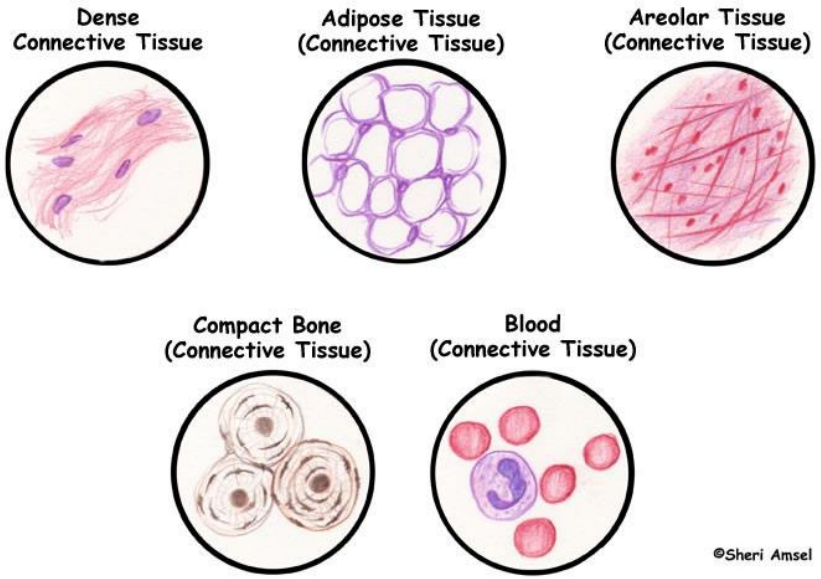


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Animal Tissue: Connective Tissue

Functions:

- Establishing a structural framework
- Transporting fluid and dissolved materials
- Protecting delicate organs
- Supporting, surrounding and interconnecting tissues
- Storing energy reserves
- Defending the body from microorganisms

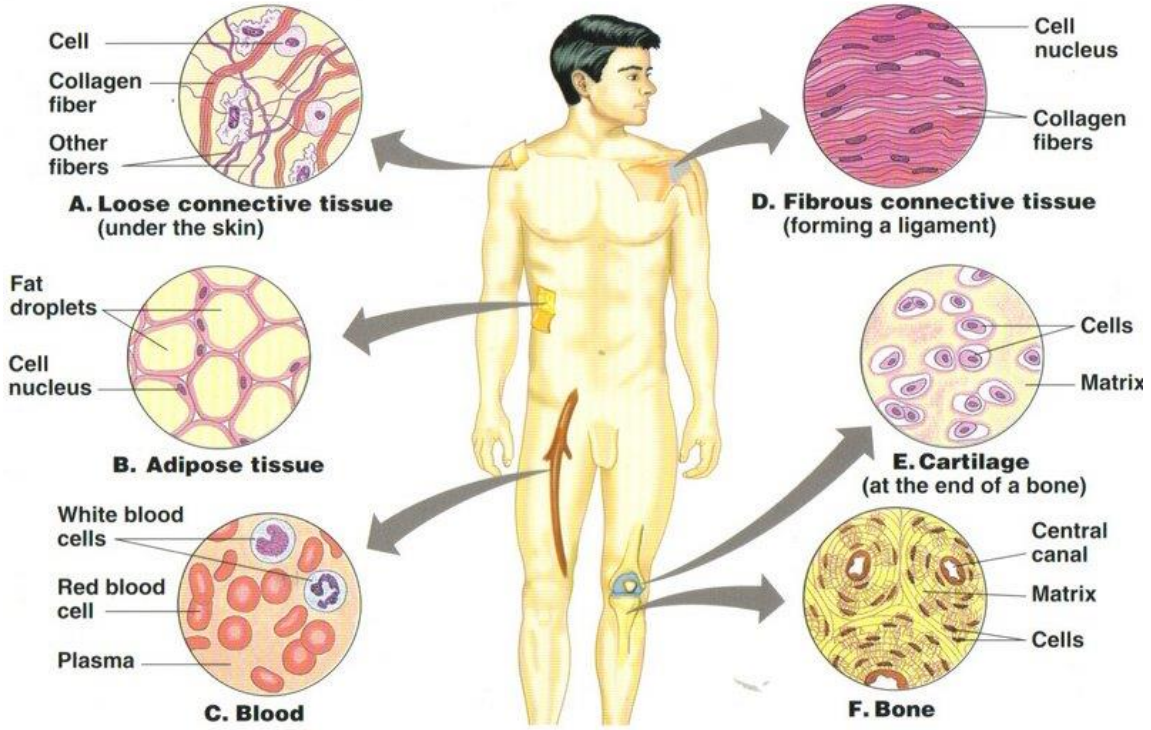


https://www.anatomynote.com/human-anatomy/cell-and-tissue/tissue-types-in-anatomy-world/

Animal Tissue: Connective Tissue

- Most abundance, widespread and varies of tissue types in the body with the widest functions
  - ❑ **Connect**, **support**, **bind**, **separate** other tissues or organs
  - ❑ Connective tissue has 3 components; **specialized cells**, **ground substance** and **protein fibers**

Fig. 20.5 Types of connective tissue



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Animal Tissue: Connective Tissue

- All types of connective tissues have three components:

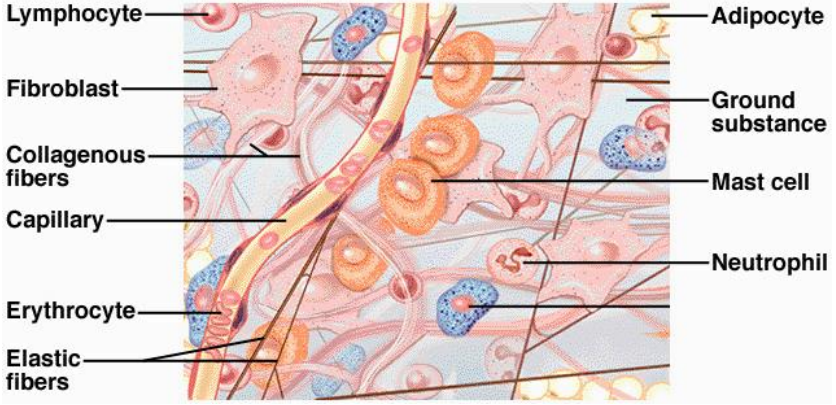
1) **Specialized cells**

2) **Ground substance** → Noncellular material, varies in textures (solid, fluid, semifluid)

3) Protein fibers

- Collagen fibers : contain collagen , provide strength and flexibility
- Reticular fibers : contain thin, highly branched collagen fibers
- Elastic fibers: elastin, not as strong as collagen, more elastic than collagen

- Ground substance + protein fibers = extracellular matrix



https://physiotherapymilan.it/paper/classification-of-connective-tissue-2544/

Animal Tissue: Connective Tissue

- Can be classified into three major categories:

Fibrous

- Have cells called fibroblast
- The cells is separated by one another by jellylike matrix containing white collagen fiber and yellow elastic fibers

Loose fibrous

Adipose tissue

Dense fibrous

Supportive

- Provide structure, shape, protection and leverage for movements

Cartilage :
- Hyaline cartilage
- Elastic cartilage
- Fibrocartilage

Bones

Fluid

- Fluid connective tissues located in blood and lymph vessels

Bloods:
- Red blood cells
- Platelets

Lymph

Lymph

Animal Tissue: Connective Tissue

- Can be classified into three major categories:

Fibrous

- Have cells called fibroblast in loose and dense and dense fibrous connective tissues
- The cells is separated by one another by jellylike matrix containing white collagen fiber and yellow elastic fibers

Loose fibrous

Adipose tissue

Dense fibrous

Supportive

- Provide structure, shape, protection and leverage for movements

Cartilage :
- Hyaline cartilage
- Elastic cartilage
- Fibrocartilage

Bones

Fluid

- Fluid connective tissues located in blood and lymph vessels

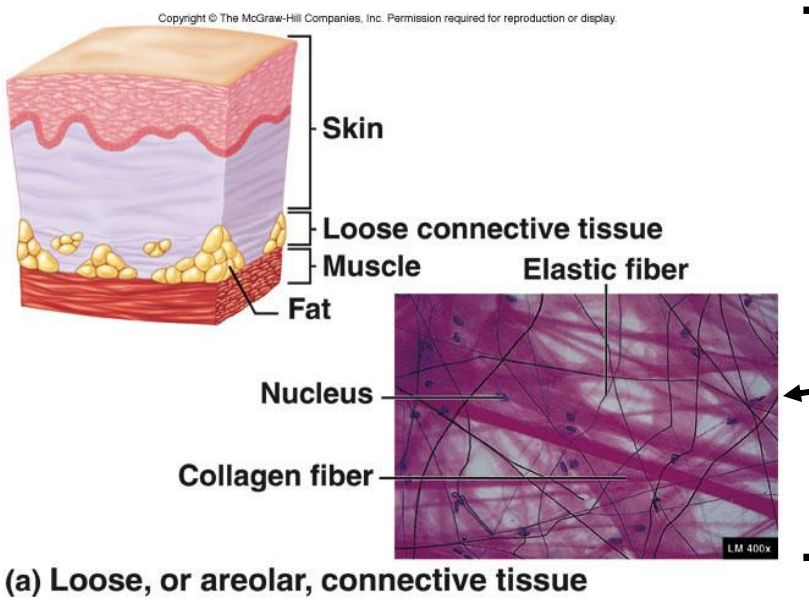
Bloods:
- Red blood cells
- Platelets

Lymph

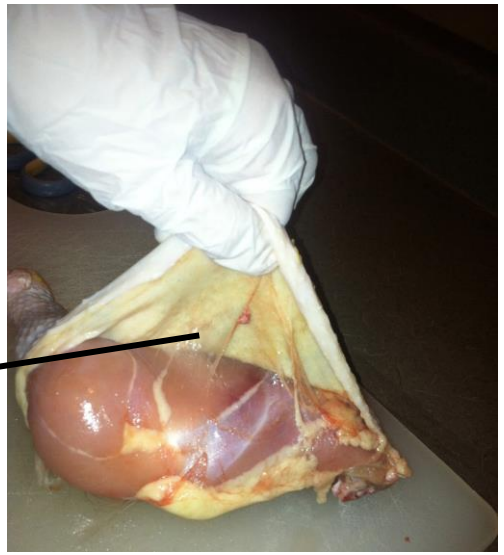
Animal Tissue: Connective Tissue

Loose Fibrous

- Location: beneath dermis of skin, digestive tract, respiratory and urinary tracts, between muscles and around blood vessels, nerves and around joints
- **Functions:**
  - ❑ Form protective covering over internal organs
  - ❑ Allow organ to expand
  - ❑ Support epithelium and many internal organs



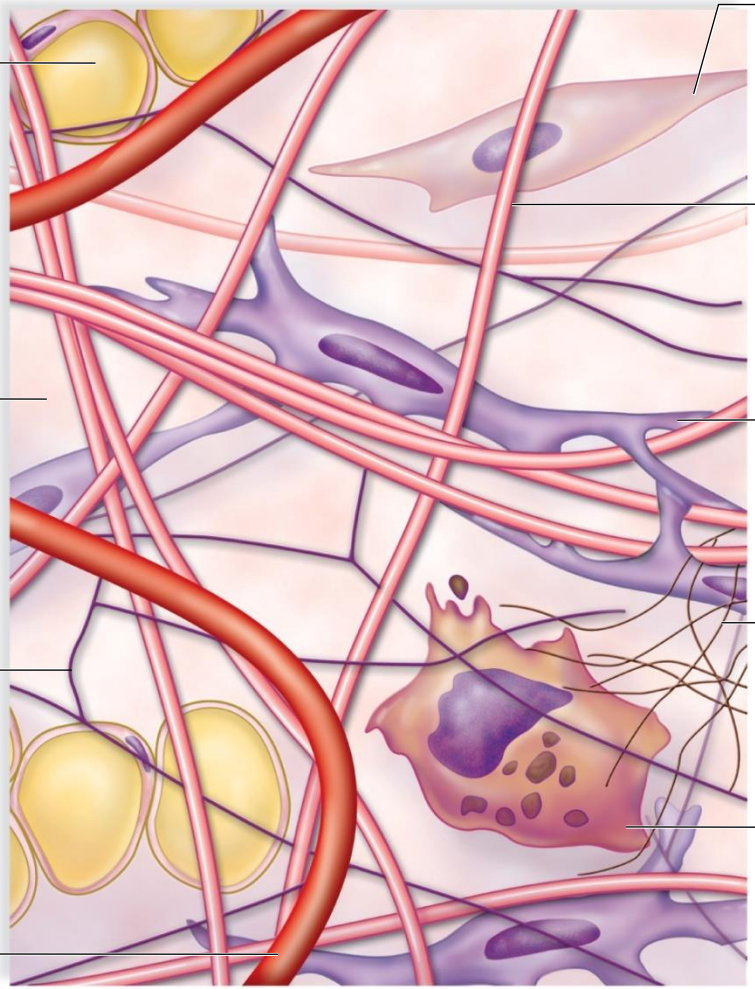
(a) Loose, or areolar, connective tissue



https://www.pinterest.com/pin/176625616612632213/

Fibrous Connective Tissue: Areolar

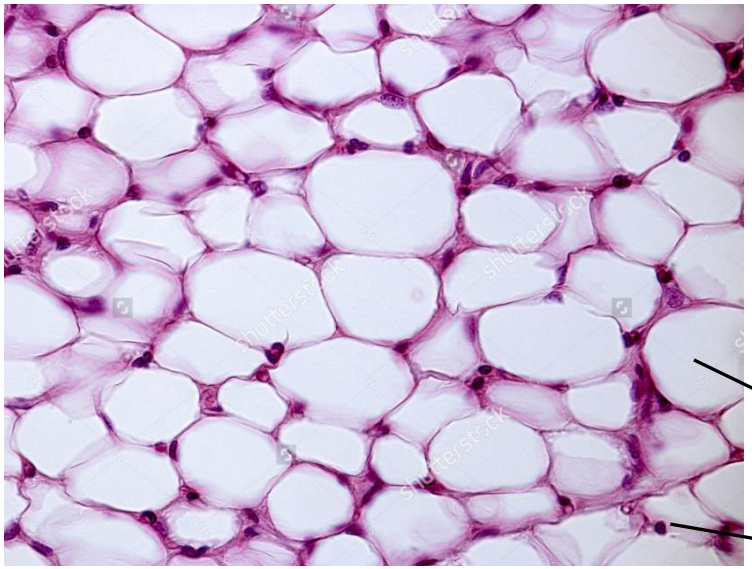
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Animal Tissue: Connective Tissue

Adipose Tissue

- Location: under skin, around kidney and eyeballs within abdomen and in breast
- **Functions :**
  - ❑ Provides reserve food fuels
  - ❑ Insulates against heat loss
  - ❑ Supports and protects organs



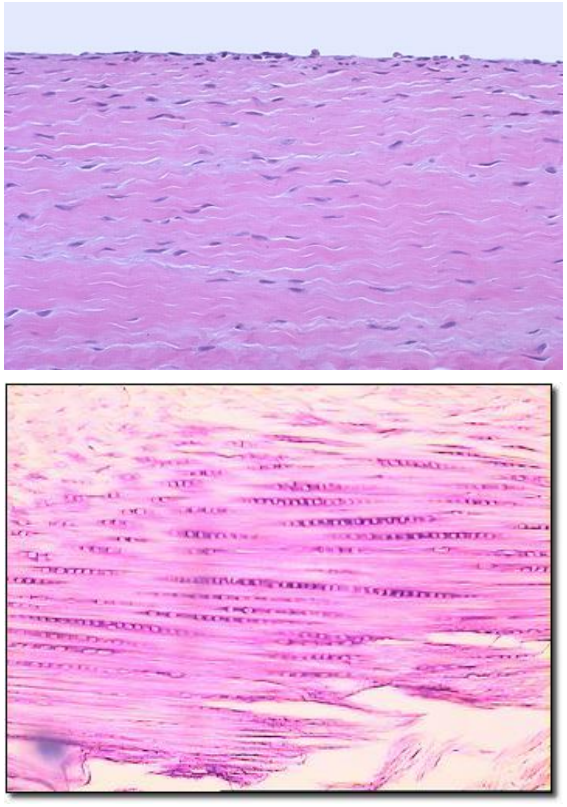
https://www.studyblue.com/notes/note/n/chapter-21-unifying-concepts-of-animal-structure-and-function/deck/5810761

- Structure : closely packed adipocytes (enlarged fibroblast that stores fat), have nucleus pushed to the side by large fat droplets

Animal Tissue: Connective Tissue

Dense Fibrous

- Strong connective tissue
- Composed of thick collagenous fibers that are pack together, a fine network of elastic fibers and few fibroblasts
- Two types of dense fibrous CT:
  - **Tendons** – connect muscle to bone
    - Less stretchy and contain fewer elastic fibers
  - **Ligaments** – connect bone to bone at joints
    - More stretchy and contains more elastic fibers



https://trello.com/c/VRq58y3F/64-no-18-dense-regular-connective-tissue-tendon-he

Animal Tissue: Connective Tissue

- Can be classified into three major categories:

Fibrous

- Have cells called fibroblast
- The cells is separated by one another by jellylike matrix containing white collagen fiber and yellow elastic fibers

Loose fibrous

Adipose tissue

Dense fibrous

Supportive

- Provide structure, shape, protection and leverage for movements

Cartilage :
- Hyaline cartilage
- Elastic cartilage
- Fibrocartilage

Bones

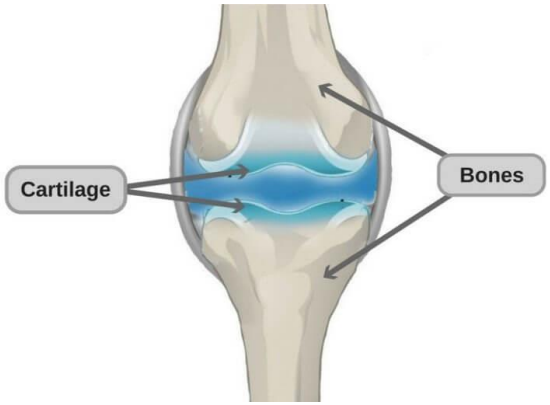
Fluid CT

- Fluid connective tissue located in blood and lymph vessels

Bloods:
- Red blood cells
- Platelets

Lymph

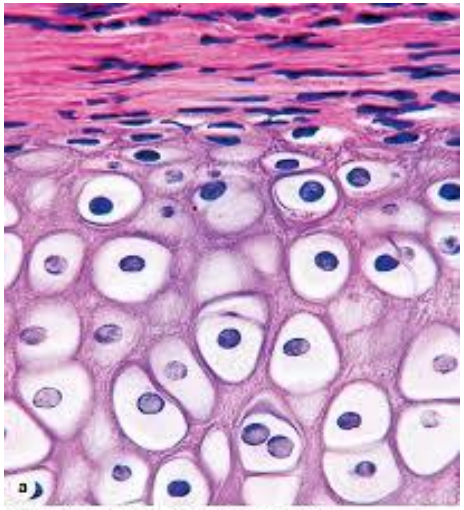
Animal Tissue: Connective Tissue



Bone	Cartilage
rigid connective tissues	Flexible connective tissues
structural protein deposited in extracellular matrix	calcium salts deposited in extracellular matrix

✓ Cartilage; cells lies in the small chambers called lacunae, separated by the matrix that is solid but flexible

1. Hyaline cartilage



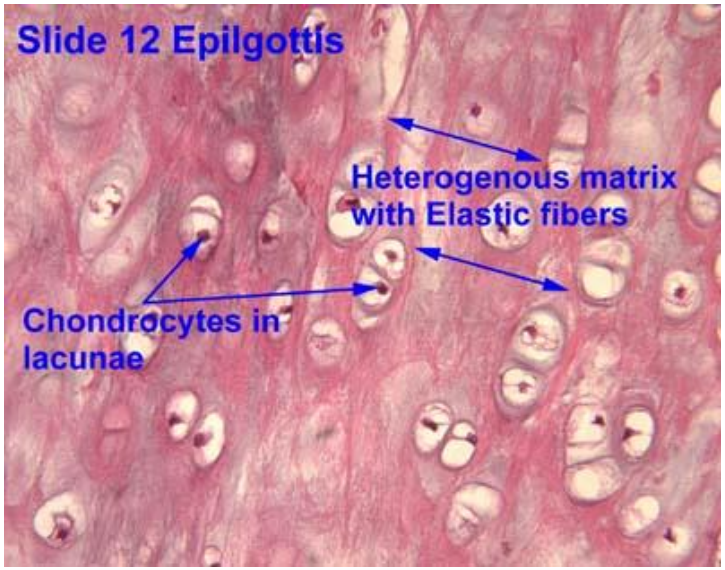
Source: Mescher AL: Junqueira's Basic Histology: Text and Atlas, 12th Edition: http://www.accessmedicine.com
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- ❑ Common types of cartilage
- ❑ Found in the nose, rings of respiratory passage, human fetal skeletal
- ❑ Widely dispersed in very fine collagen fibers
- ❑ The weakest type of cartilage

Animal Tissue: Connective Tissue

2. Elastic cartilage

- Has more elastic than hyaline cartilage (more flexible)
- Found in outer ear, Eustachian tube and epiglottis
- Provides strength, elasticity and maintains shape of certain structure



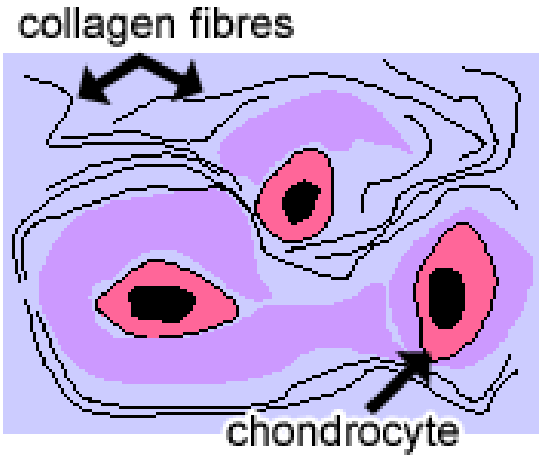
https://www.ouhsc.edu/histology/Text%20Sections/Cartilage.html

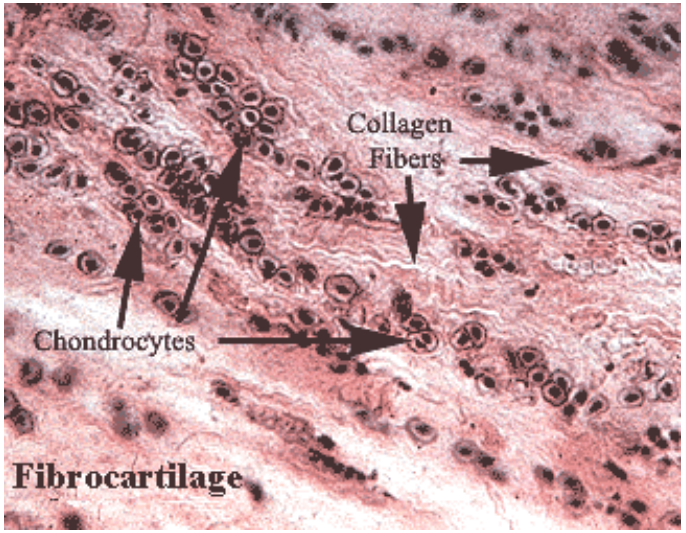
- Chondrocytes are found in a threadlike network of elastic fibers within the matrix

Animal Tissue: Connective Tissue

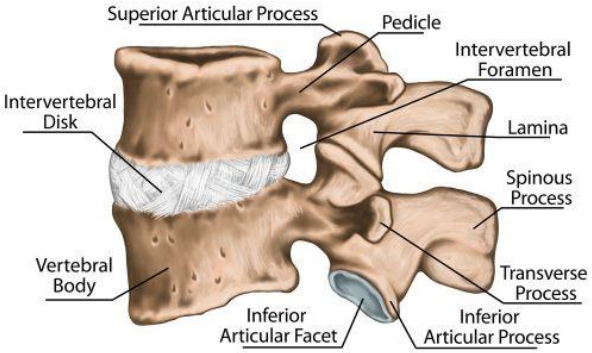
3. Fibrocartilage

- Strongest kinds of cartilage
- Has matrix containing strong collagen fibers
- Found in the area that must withstand tension and pressure : pad between vertebrae, wedges in the knee joint





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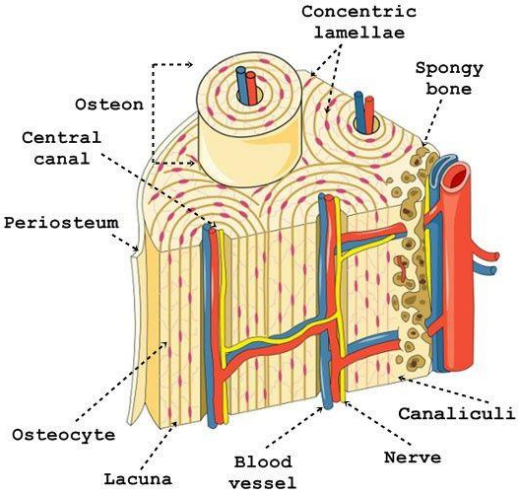


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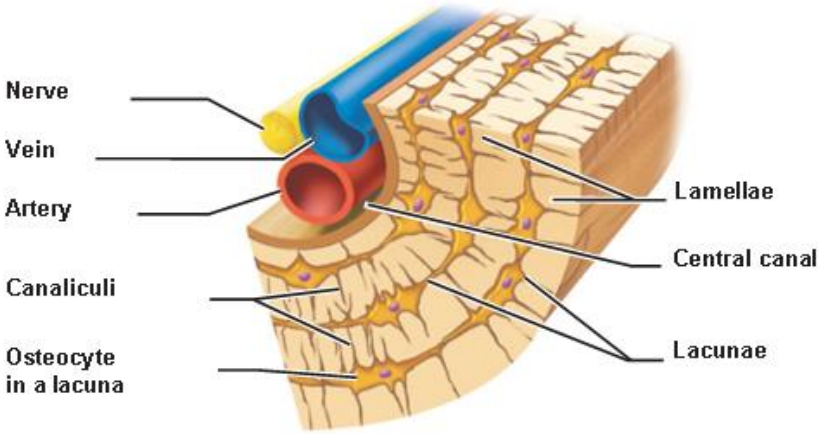
Animal Tissue: Connective Tissue

Bone

- Ground substance contains hard matrix of inorganic salts (mainly calcium salts) deposited around protein fibers
  - ❑ **calcium salts** : give rigidity
  - ❑ **protein fibers** : provide elasticity and strength
- Bone cells (**osteocytes**) – located in lacuna that the cells are responsible for maintaining the bone matrix
- Lacuna – arranged in concentric circles within osteon around tiny tubes (central canal/ Harvesian canal)



http://encyclopedia.lubopitko-bg.com/Bone.html



https://antranik.org/cartilage-and-bones/

Animal Tissue: Connective Tissue

- Can be classified into three major categories:

Fibrous

- Have cells called fibroblast
- The cells is separated by one another by jellylike matrix containing white collagen fiber and yellow elastic fibers

Loose fibrous

Adipose tissue

Dense fibrous

Supportive

- Provide structure, shape, protection and leverage for movements

Cartilage :
- Hyaline cartilage
- Elastic cartilage
- Fibrocartilage

Bones

Fluid

- Fluid connective tissue located in blood and lymph vessels

Bloods:
- RBC and WBC
- Platelets

Lymph

Animal Tissue: Connective Tissue

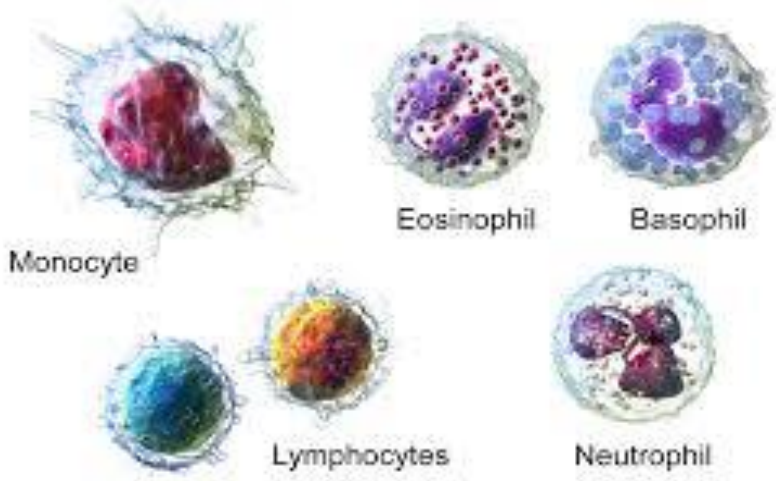
Blood

- Connective tissues that are embedded in a liquid matrix (plasma)
- **Functions**:
  - ❑ Transport nutrients and oxygen to cells
  - ❑ Removes CO_2 and wastes
  - ❑ Ph, ions and PH balance
  - ❑ Distributes heat

Two types:



Red blood cells:
carry oxygen

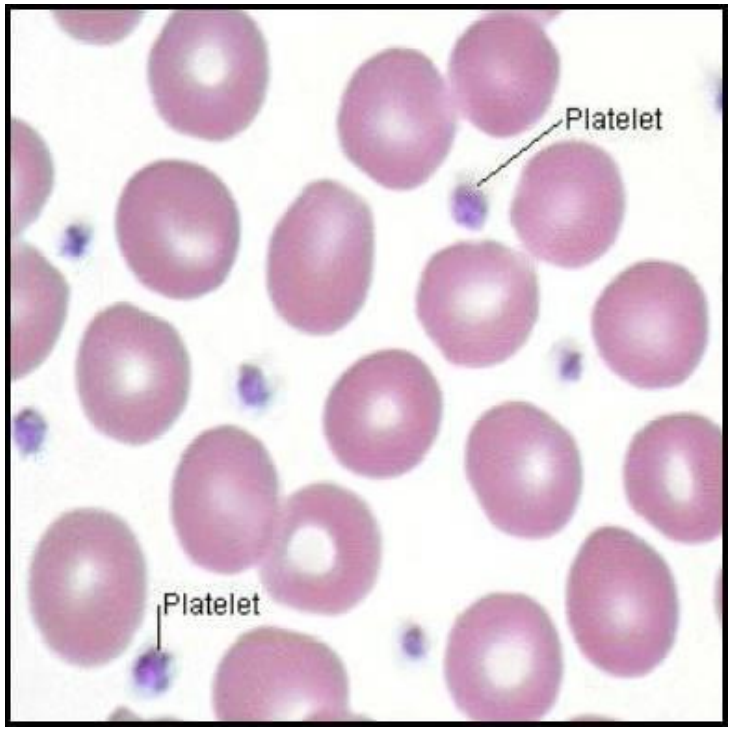


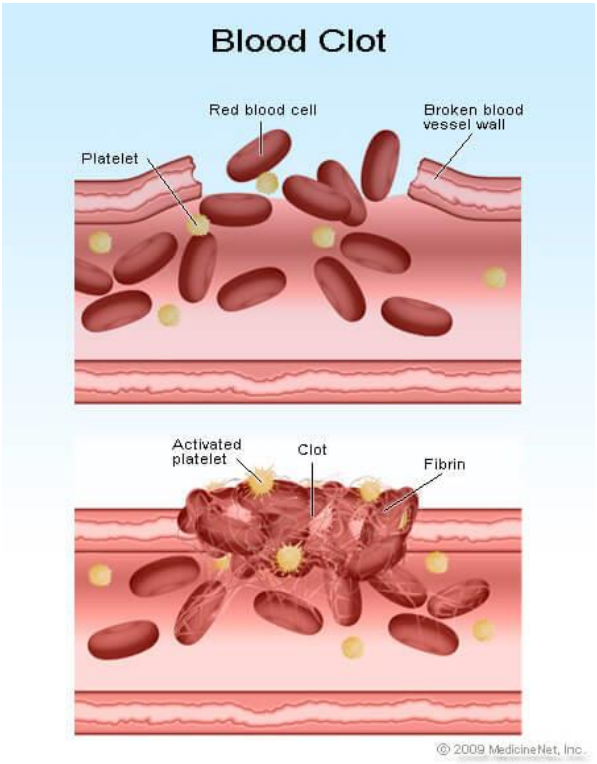
White blood cells:
fight infection

Animal Tissue: Connective Tissue

Platelet

- Incomplete cells
- Fragments of large cells; present only in bone marrow
- When blood vessel is damaged – forms a plug that seal the vessels
- Assists the blood clotting process

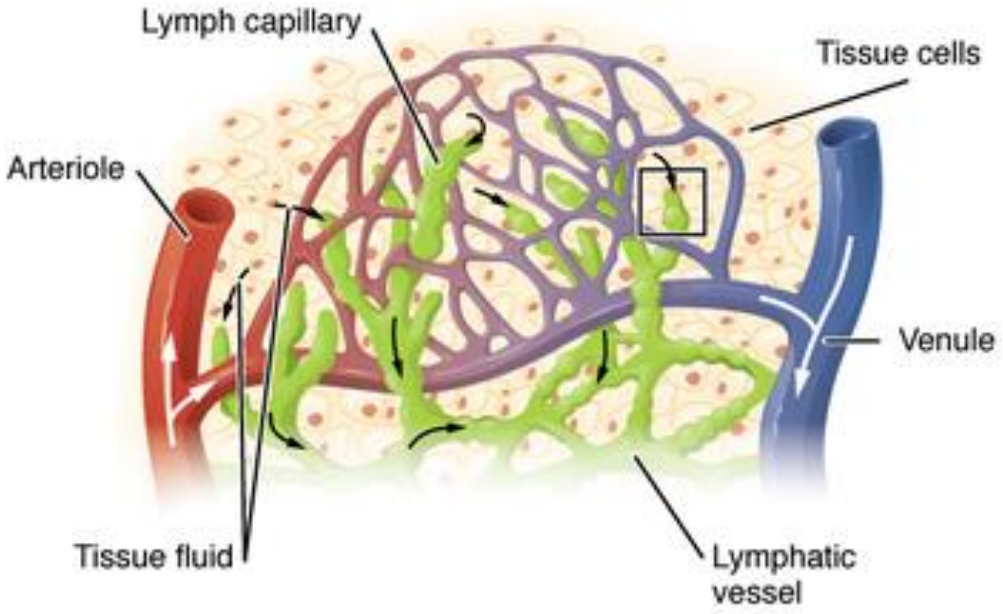




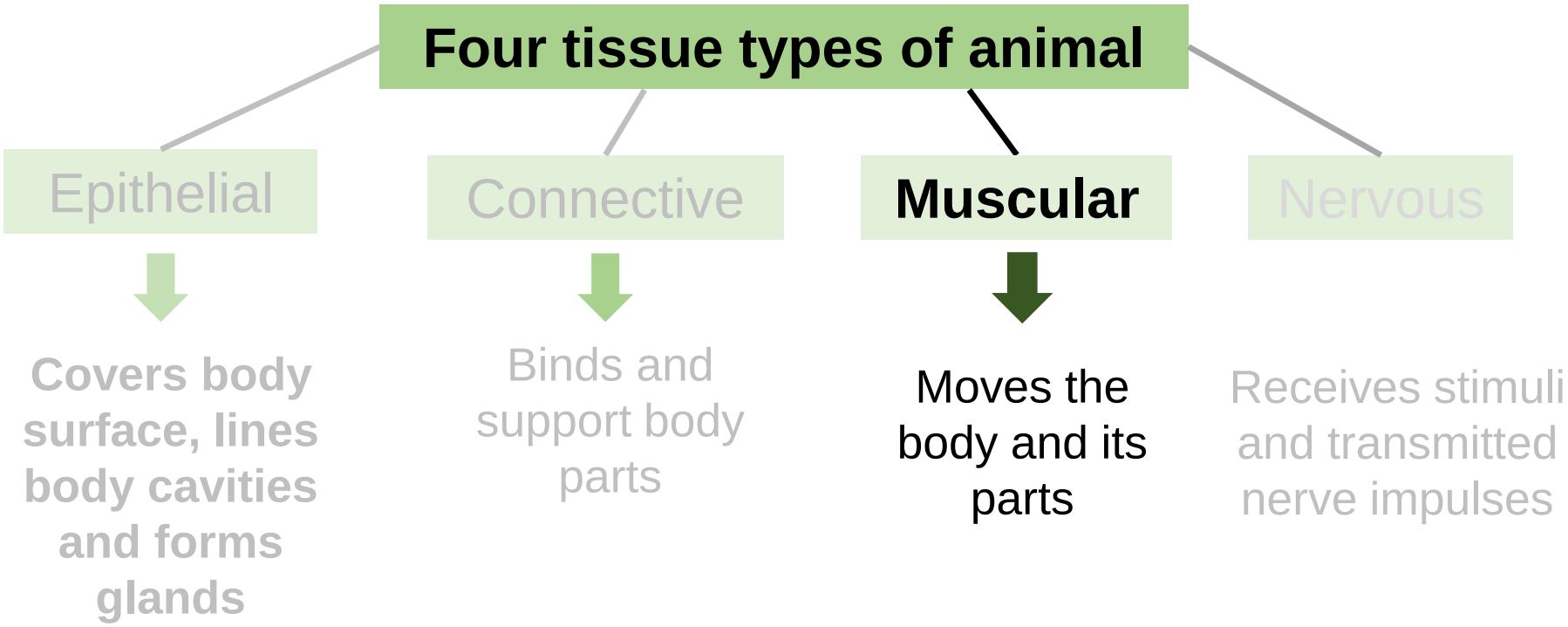
Animal Tissue: Connective Tissue

Lymph

- A fluid connective tissue in lymphatic vessels
  - ❑ **Absorb interstitial fluid** and return it into cardiovascular systems
  - ❑ Have lacteal that **absorb fat molecules** from small intestine
- Lymph contains lymphocytes that remove foreign material from lymph vessels via lymph nodes



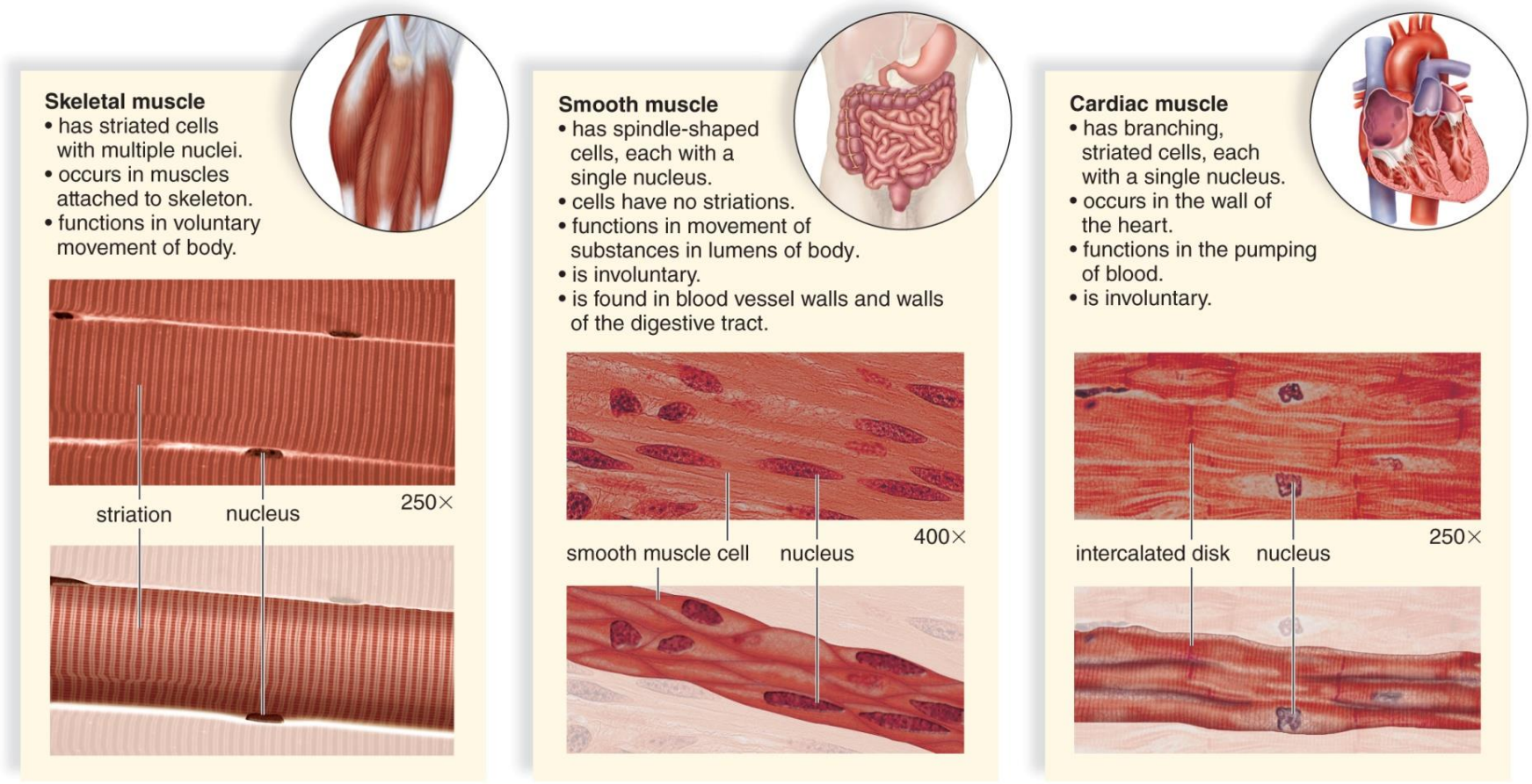
Animal Tissue: Muscular Tissue



Animal Tissue: Muscular Tissue

- Composed of cells that are composed of **muscle fibers**
- Three types of muscles tissues:

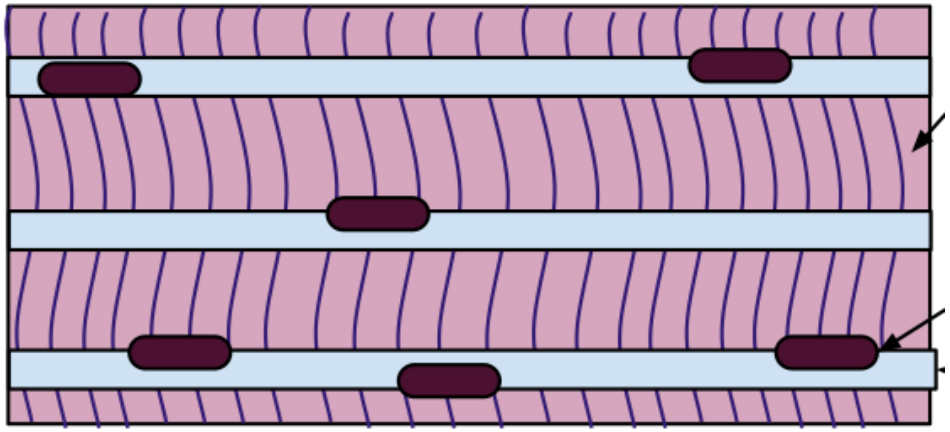
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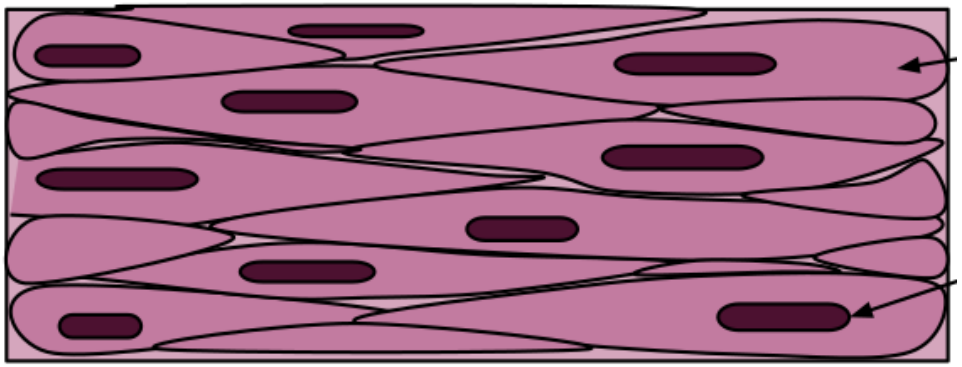


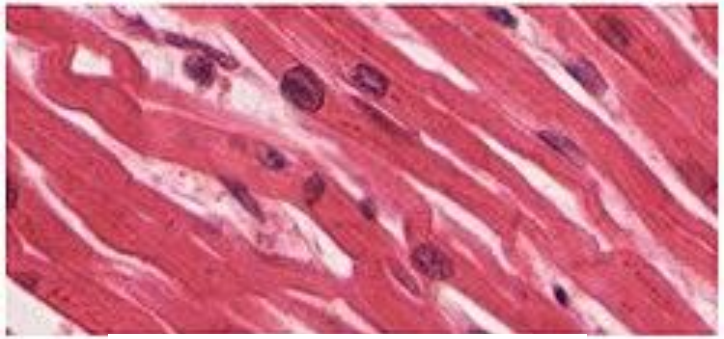
Skeletal Muscle



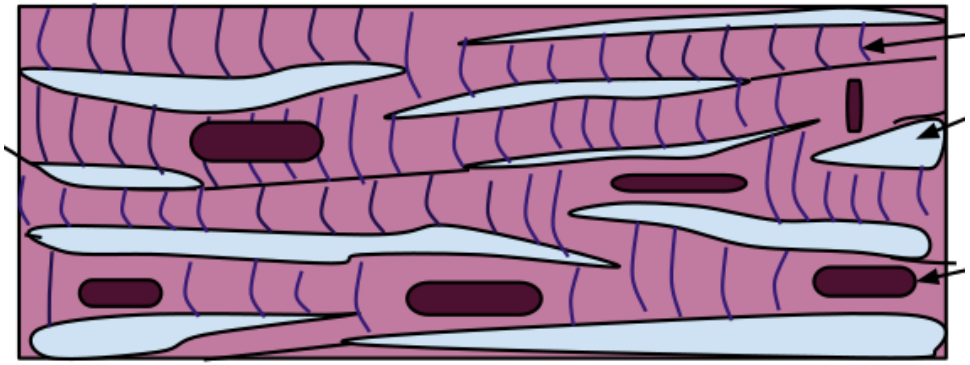


Smooth Muscle

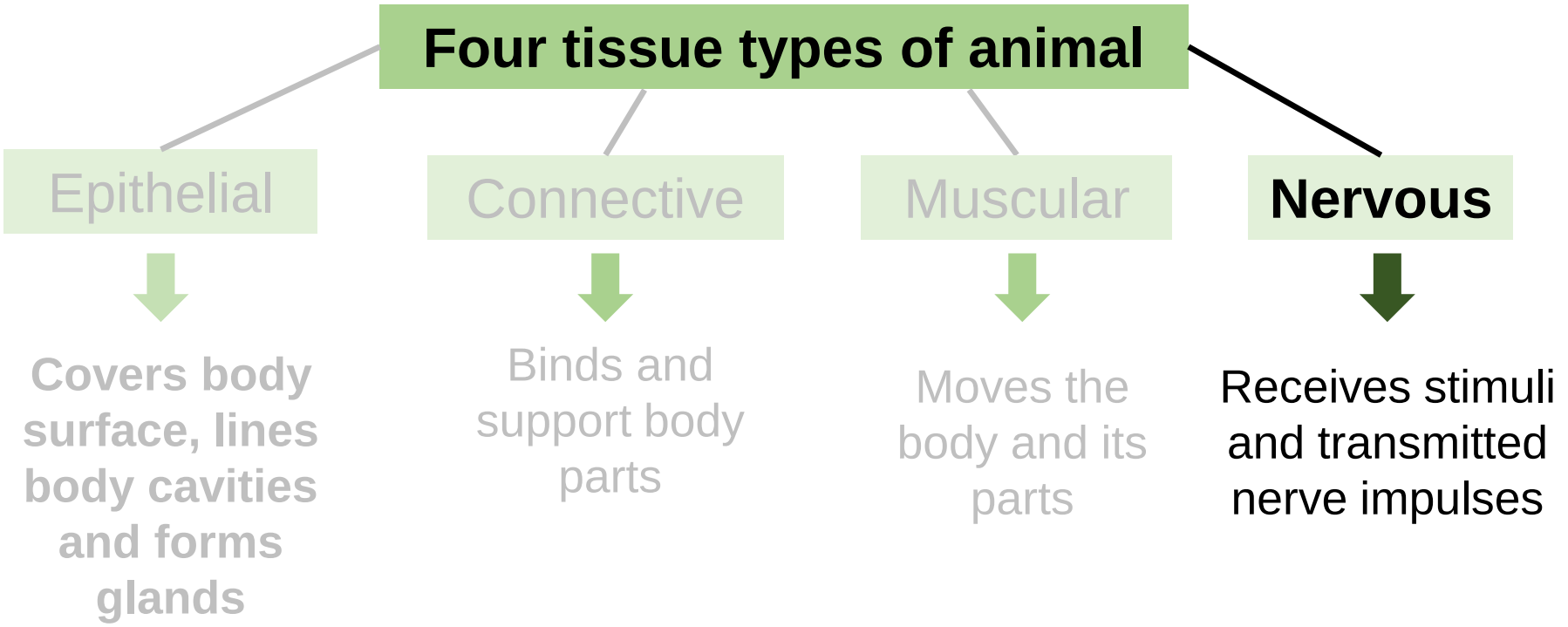




Cardiac Muscle

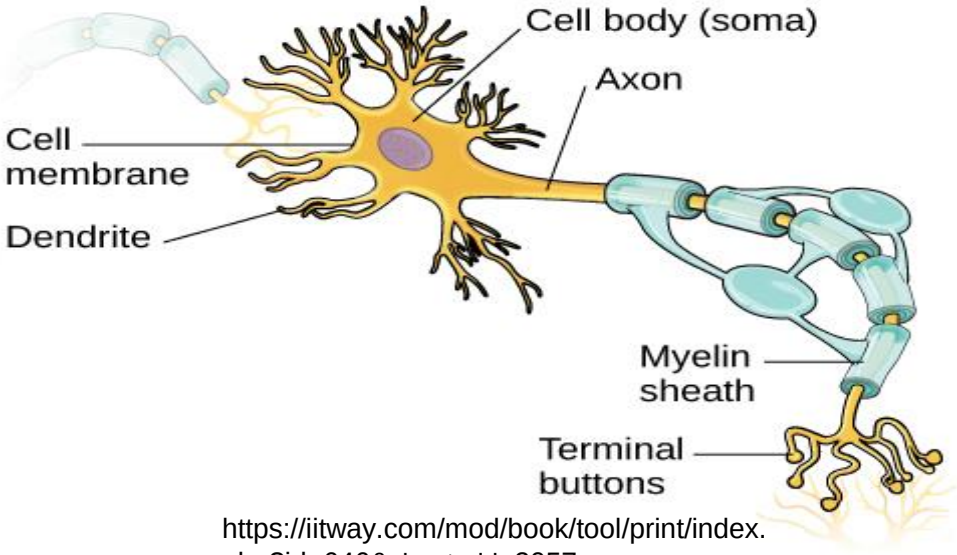


Animal Tissue: Nervous Tissue

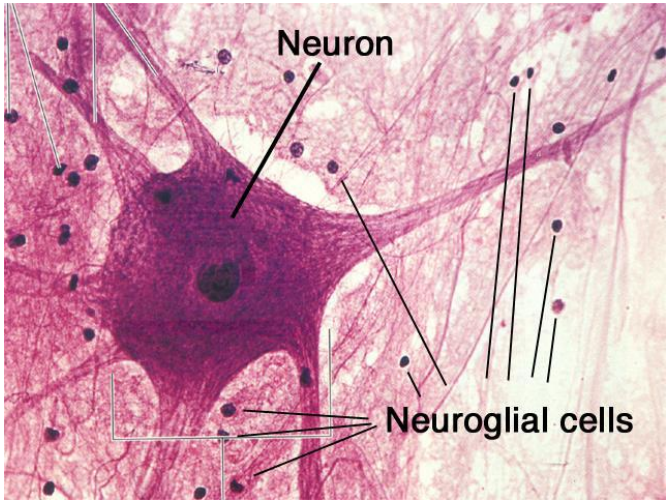


Animal Tissue: Nervous Tissue

- Contains nerve cells called neurons and neuroglia (supporting cells)



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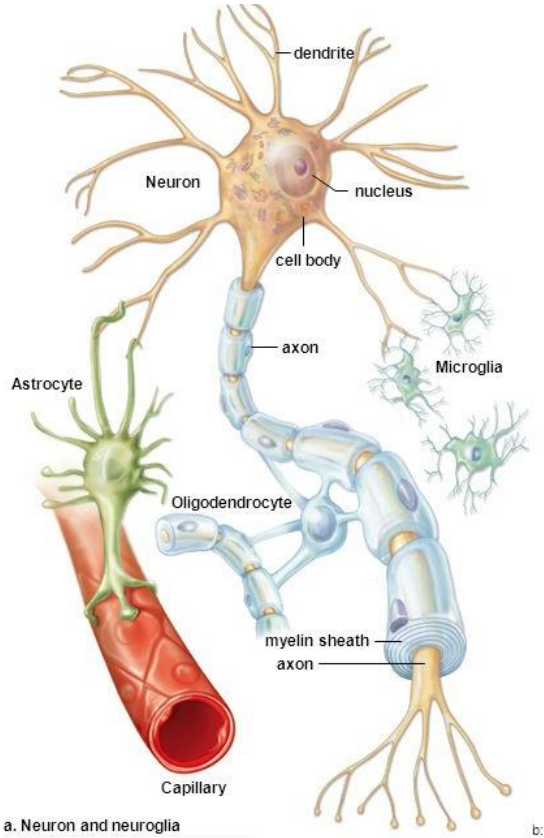


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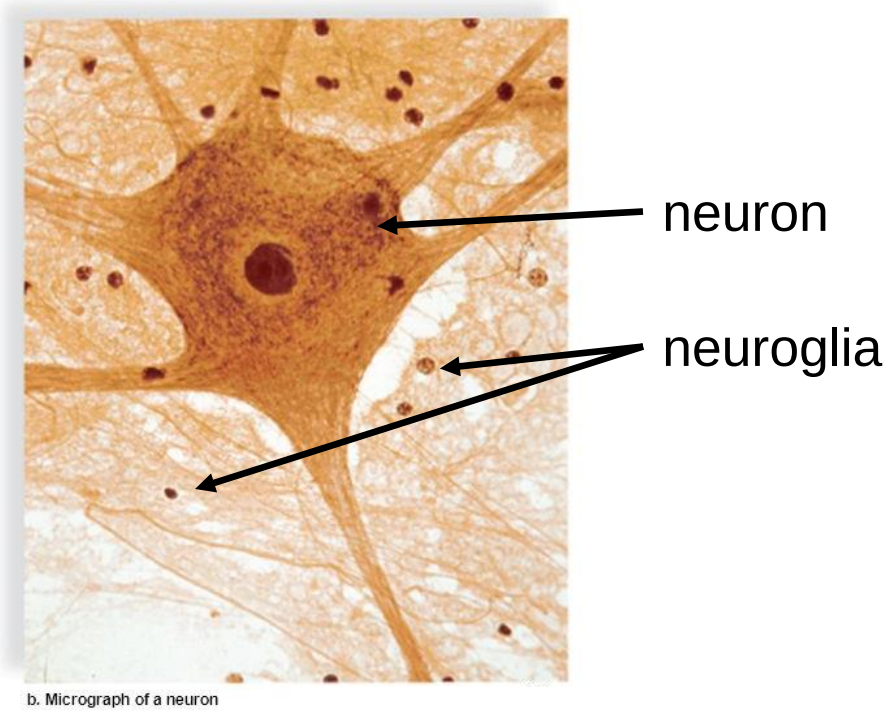
- Neurons has three parts :
  - ❑ Dendrites: receives stimulus and conducts signals to the cell body
  - ❑ Cell body has most of the cytoplasm and contain nucleus
  - ❑ Axon : conducts nerves impulse away from the cell body and its cover by the myelin sheath

Animal Tissue: Nervous Tissue

- **Neuroglia** make up half the volume of the brain
- **Function:** support and nourish neurons
- Several types of neuroglia found in the brain: **microglia**, **astrocytes** and **oligodendrocytes**



a. Neuron and neuroglia



b. Micrograph of a neuron

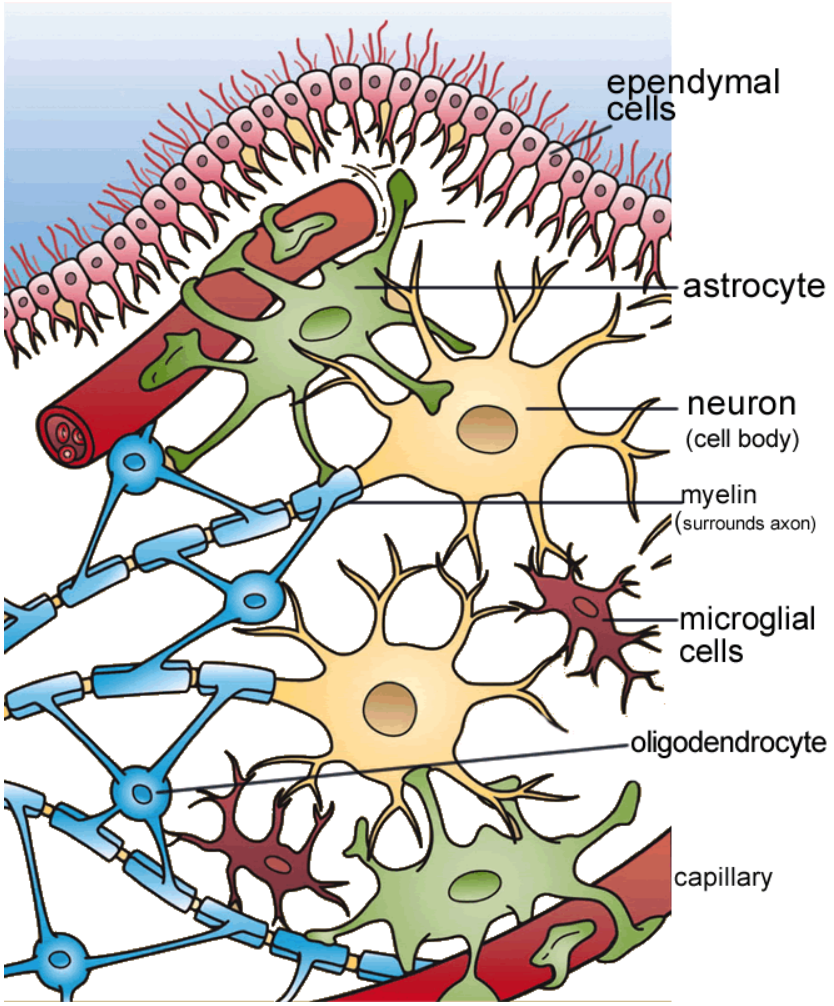
b: © Ed Reschke

Animal Tissue: Nervous Tissue

Microglia – support cell neurons and phagocytize bacteria and cellular debris

Astrocytes – provide nutrient and produce hormone known as cell derived growth factor

Oligodendrocytes – form myelin in the brain



http://anatomycorner.com/main/2015/11/25/neuroglial-cells/












THANK YOU





